

0. Summary

The continued expansion of variable renewable generation, the electrification of heat, mobility and industry, growing network congestion, and the progressive differentiation of trading and system-service markets are fundamentally reshaping the logic by which energy assets are valued. For developers and operators of PV-and-BESS, wind-and-BESS, and hydrogen-and-BESS portfolios, project quality is now determined by a broader set of conditions than resource excellence, investment discipline, grid connection, and conventional marketability. Increasingly decisive is whether an asset or portfolio has been designed, contractually structured, and operated in a way that preserves its degrees of freedom – and keeps them economically accessible – across changing market, network, and operating conditions.

This is the operational meaning of flexibility. It is a usable capacity to change behaviour deliberately, embedded within a robust commercial and operational framework – distinct from storage and distinct from any single market product. A degree of freedom acquires strategic relevance when it is controllable, contractually compatible, cleanly attributable in balancing terms, and operationally governable. The significance of this shift is visible in hard numbers. In Germany, redispatch costs reached approximately €590 million in 2021, against around €240 million the previous year; by 2024, figures cited in current transmission system operator and Bundesnetzagentur documentation point to expenditure of €2.7 billion. At the European level, demand-side flexibility for 2030 carries an estimated additional potential of 164 GW of upward flexible power and 130 GW of downward flexible power, associated with modelled savings of €4.6 billion in generation costs, 15.5 TWh less curtailment, 37.5 Mt of avoided CO₂ emissions, and between €11.1 and €29.1 billion per year in reduced distribution network investment. Flexibility thus constitutes a response to a coordination challenge that is visibly becoming both costlier and more complex.

This is particularly evident in hybrid portfolios. PV, wind, BESS, electrolysis, hydro, heat pumps, electric vehicle charging infrastructure, and data centres each open different degrees of freedom – differing in power, duration, response speed, state-dependence, locational relevance, and integration capability. Their value arises from the quality of their coupling rather than from simple addition. What matters is whether a portfolio possesses these degrees of freedom and can rank and exercise them in an orderly fashion across different market and operating conditions.

A changed view of markets follows from this. Day-ahead and intraday markets organise large-area price formation and schedule adjustment. Balancing energy markets address standardised short-term system flexibility. Local and vertical flexibility markets access, in addition, those degrees of freedom whose value depends substantially on location within the network, the direction of activation, and the coordination between network, balancing group, aggregation, and platform logic. These layers form different coordination strata of the same system rather than competing alternatives. Their robustness depends on appropriate participation incentives, adequate liquidity, precise prequalification, reliable activation and settlement logic, and a regulatory framework capable of translating technical potential into institutional practice.

For developers, operators, IPPs, aggregators, and integrated portfolio actors, the design logic shifts accordingly. A project is a building block of a future action space – one in which asset characteristics, network situation, contract structure, market access, balancing group logic, data availability, and operating model interlock. The true quality of a portfolio reveals itself in the discipline with which its degrees of freedom are preserved, valued, and managed under uncertainty. Current practice adds a further dimension: flexibility is already being filtered at the point of grid connection, where high connection demand, constrained capacity, and maturity-based prioritisation make network and system benefit an increasingly independent criterion for assessing storage and flexibility projects.

The compendium develops this logic across three equally weighted layers. The Asset and Engineering Layer examines which degrees of freedom the relevant asset classes genuinely open up, and which constraints govern their hybridisation. The Market and Contract Layer analyses wholesale markets, balancing energy, local and vertical flexibility markets, direct marketing, route-to-market, and PPA, CPPA, and CfD structures as the economic and institutional spaces in which value is realised. The Operating and Portfolio Layer addresses forecasting, scheduling, dispatch, hedging, ETRM, balancing group management, settlement, governance, and organisational design. Together, these layers constitute an integrated decision architecture for the development, marketing, and operational governance of flexible renewable portfolios – a purposeful alternative to yet another market survey.

The perspective throughout is deliberately technical, operational, and commercial at once. Flexibility is an expression of coordinated optionality in an energy system that demands new forms of operational governance under volatility, network constraint, decarbonisation pressure, and growing data intensity. In the Ariadne scenario context, Germany's 2045 energy system is projected to comprise around 940 GW of installed generating capacity – including 300 GW of wind and 440 GW of photovoltaics – complemented by flexible backup gas capacity and growing roles for storage and electrolyzers. The strategic significance of flexibility lies precisely here: in the ability to translate physical degrees of freedom into durable action spaces, and to sustain those spaces across markets, contracts, and operating models.

The analysis that follows draws on a perspective that treats technical development, market logic, and operational control as inseparable. It is grounded in more than twenty-five years of experience in the energy industry, including executive responsibility for building and operating a European and Canadian flexibility market platform for capacity and energy.

1. Strategic context

1.1 The shift in value logic

The European energy industry is passing through a phase in which value creation is shaped by a broader set of conditions than installed capacity, resource quality, and conventional marketability. As volatile renewable generation expands, further consumption sectors are electrified, network congestion becomes a structural operating reality, and the differentiation of market and system-service mechanisms deepens, the focus shifts to those degrees of freedom that a portfolio can exercise under real constraints.

The assessment of assets changes with it. A project must be judged on whether its technical, contractual, and operational structure preserves the degrees of freedom that remain economically usable across shifting price, network, and state conditions. This is the starting point from which flexibility derives its operational meaning.

1.2 Decarbonisation, electrification and system densification

Decarbonisation and electrification increase the share of renewable generation and, simultaneously, densify the system. More weather-dependent infeed, more electricity-based heat, more electric mobility, new hydrogen pathways, rising data centre loads, and growing demands on network and system stability all increase the number of simultaneous interactions within the system. This densification carries a measurable economic dimension. In Germany, redispatch costs stood at approximately €590 million in 2021, against around €240 million in 2020; by 2024, figures cited by the four transmission system operators and the Bundesnetzagentur point to redispatch expenditure of €2.7 billion. At the European level, the benefits of activated demand-side flexibility for 2030 are quantified as including 164 GW of upward flexible power, 130 GW of downward flexible power, €4.6 billion in lower generation costs, 15.5 TWh less curtailment, and between €11.1 and €29.1 billion per year in reduced distribution network investment. The case for flexibility therefore rests on the expansion of renewable generation and, with equal force, on a coordination challenge that is becoming measurably costlier and more complex.

This is a structural shift. It concerns the technical architecture of assets, the design of portfolios, the logic of contracts, and the quality of operational management. Where degrees of freedom are preserved, new possibilities for value creation and system stabilisation emerge. Where they are foreclosed early – through contractual, technical, or organisational choices – dependence on external balancing mechanisms grows correspondingly.

This densification is concentrated, moreover, in the distribution networks. The greater part of new renewable generation, together with a rising number of large point loads – data centres, battery storage, and large heat pumps – connects at distribution level, which is increasingly the place where the transition is physically realised. As connection requests accumulate against constrained capacity and long reinforcement lead times, network and system benefit becomes an increasingly independent criterion for assessing a flexibility project, alongside its market value.

1.3 Flexibility as a response to volatility and congestion

Volatility generates uncertainty, profile displacement, price dispersion, and congestion pressure. Where a portfolio can respond deliberately to these conditions, volatility becomes a usable action space. Flexibility is the capacity to affect this response – an answer to the widening gap between rigid infrastructure and dynamic system reality, rather than an abstract system property.

A structural information and coordination problem reinforces this framing. Under current market and network tariff logic, network users are in principle exposed to the same exchange price signals and predominantly static network charges; local congestion situations are therefore only partially translated into economically effective signals. Current material published by Germany's four transmission system operators makes this point explicitly: where local price differences and location-specific signals are absent, incentives for congestion-avoiding behaviour are correspondingly weak, and network bottlenecks are corrected ex post through redispatch. Additional flexibility, moreover, does not automatically act in a system-beneficial manner without appropriate network- and location-specific signals. Flexibility thus becomes a more precise form of system coordination under conditions of local information scarcity, alongside its role as a response to volatility.

Flexible loads, conversion plant, hybrid structures, local control architectures, and contractually preserved mobility can each serve the same systemic purpose as storage, through different means. The compendium follows this broader logic throughout: degrees of freedom as a designable architecture, rather than individual technologies examined in isolation.

1.4 The electricity market as a multilayered coordination order

The electricity market is a multilayered coordination order, one that reflects underlying physical reality only in part. Day-ahead and intraday markets organise large-area price formation and near-term schedule adjustment. Balancing energy markets provide standardised short-term system stability. Local and vertical flexibility markets access those values arising from locational specificity, network state, direction of activation, and institutional coupling. These layers are complementary coordination strata of the same system. Reading markets well therefore means attending to the superposition of different market and activation logics acting on the same degree of freedom. Value is created where this superposition is understood, ranked, and operationally governed. Local flexibility markets demonstrate with clarity that product design, prequalification, activation logic, data access, settlement, and the allocation of roles and responsibilities are preconditions for functional market integration rather than secondary details. The regulatory framework for flexibility is, moreover, not static. The European Electricity Market Design Reform, RED III, and further battery, data, cybersecurity, and network tariff regulation are each gaining in significance, altering product requirements, permitting conditions, evidence structures, and the future embedding of non-fossil flexibility within market and network architectures.

1.5 Data, digitalisation and operational controllability

As system density grows, data, forecasting, and controllability grow in importance with it. Digital infrastructure is the precondition for states becoming visible, degrees of freedom becoming assessable, and operational decisions becoming robust. Without telemetry, state awareness, forecasting, scheduling, and dispatch, flexibility remains latent – present technically, yet unavailable as an economically usable action space.

Digital capability is therefore inseparable from the asset world. It connects engineering, market participation, and portfolio management within a common logic. This is why the compendium treats flexibility as an architecture of operational controllability, alongside its market and engineering dimensions.

1.6 Local and vertical flexibility markets as a complementary model

Local and vertical flexibility markets complement the wholesale market and balancing energy rather than displacing either. They add a coordination layer in which locational specificity, network state, and concrete activation effect become economically relevant. Their strength lies precisely in this function; so too do their design requirements. Local markets must be sufficiently standardised to enable participation, scaling, and liquidity, without sacrificing their locational purpose.

The distinction between market layers can be stated plainly. A zonal wholesale market answers the question of at what price energy is traded within or between bidding zones. A local flexibility market answers the question of what value a change in generation or consumption at a specific network node or within a specific network section carries. These two questions are related and irreducible to one another. In a coupled day-ahead market, a price can be economically consistent and simultaneously insufficient to resolve a local bottleneck.

Vertical flexibility markets become relevant where different system levels must be coordinated: local loads and generators, aggregators, balancing groups, distribution system operators, transmission system operators, and, where applicable, platform operators. The denser this interaction, the more important becomes an architecture in which responsibilities, activation rights, remuneration, and data flows are coherently aligned. This market form is an expression of an energy reality in which large-area price formation alone does not transmit all physically and economically relevant signals with sufficient resolution – a practical instrument rather than an ideological proposition.

1.7 Scope and deliberate boundaries

This compendium examines flexibility from the perspective of asset design, market access, contract structure, portfolio management, and operational controllability. Its central question is how technical degrees of freedom in renewable and hybrid portfolios can be designed, contractually structured, marketed, and operated to produce durable economic and systemic action spaces.

The scope is integrative by design. Flexibility is treated as a set of conditions that emerges at the interfaces: between engineering and marketing, between contract logic and dispatch, between balancing group responsibility and activation capability, between state awareness, platform logic, validation, and settlement. The compendium follows a connected systems perspective in which degrees of freedom become relevant when they are technically available, contractually compatible, balancing-consistent, and operationally governable.

The focus falls on flexibility-capable renewable and hybrid portfolios – in particular PV-and-BESS, wind-and-BESS, and hydrogen-and-BESS configurations – as well as further relevant flexibility resources including hydro, heat pumps, EV charging infrastructure, and data centres. Asset and engineering logic, market and contract logic, and operating and portfolio logic are each examined with the aim of rendering intelligible the translation of physical optionality into robust value architectures.

The boundaries are deliberately drawn. This compendium is not a comprehensive legal commentary, a country-complete regulatory manual for all European flexibility markets, a specialised trading handbook for advanced utility trading floors, or a monographic treatment of individual technologies. Where deeper mathematical, regulatory, or technology-specific grounding is required, the annexes and reference base fulfil that function.

Its purpose is a different one: to establish a robust ordering framework within which technical degrees of freedom, market mechanisms, contract structures, and operational management logics become jointly legible. It is addressed to developers, operators, IPPs, aggregators, portfolio managers, and strategic decision-makers who wish to understand and shape flexibility as an integrated architecture of optionality, risk, and controllability.

2. Conceptual and functional foundations

2.1 Why terminological precision is productive here

Flexibility is one of those terms that commands broad agreement in energy industry discourse whilst meaning different things to different interlocutors. So long as the conversation remains abstract, this ambiguity carries little consequence. The moment asset design, marketing structures, balancing group logic, platform architecture, or portfolio management become concrete, however, imprecise language begins to obstruct rather than assist. At that point, the choice of terms determines whether a matter is grasped precisely or merely circled rhetorically.

This compendium does not pursue lexical completeness. It establishes a small set of load-bearing concepts with sufficient rigour that technical, market, and operational propositions remain mutually coherent throughout. The main text leads; a glossary does not substitute for it. What carries the argument remains in the running prose; supplementary definitions belong in the annex. Terminological precision here is a condition of decision-making capacity rather than a matter of style. It creates a shared reference space across engineering, marketing, balancing, portfolio management, and operating model design – indispensable in a field where the same physical reality can read very differently depending on market role, contract structure, or operating state.

2.2 Flexibility, degrees of freedom, and coordinated optionality

In this compendium, flexibility denotes the usable capacity of an asset or portfolio to change its injection, withdrawal, storage, or energy conversion deliberately, as a function of time, location, state, and signal. The operative word is usable. Technical potential alone is insufficient. A degree of freedom acquires economic and systemic relevance when it is observable, controllable, contractually permissible, compatible with balancing obligations, and operationally governable.

The degree of freedom is the more concrete category. It describes the actual room within which an asset or portfolio can behave differently from its undisturbed or purely passive operation. This room may lie on the generation side, the demand side, in storage, or in conversion. It may be fast or slow to respond, locally or system-wide relevant, brief or sustained in its useful duration. Its quality is measured by its robustness under real operating constraints rather than by its theoretical existence.

Three levels of application can usefully be distinguished. Own-use denotes the deployment of a degree of freedom within the operator's own sphere – for cost reduction, self-consumption optimisation, or profile smoothing. Network-oriented application targets the influence of local or regional network conditions. System-oriented application addresses the system as a whole, the power balance, frequency response, or large-area market and schedule logic. These levels are analytically separable; in practice they are frequently intertwined.

To capture this intertwining precisely, the compendium works with the concept of coordinated optionality. The term does not describe the additive exploitation of arbitrary technical potentials, but the disciplined preservation of action options. This optionality has four interconnected dimensions: a physical dimension, since every asset operates within real technical limits; a market dimension, since not every degree of freedom is realisable in every product or time window; a contractual dimension, since contracts either preserve or curtail degrees of freedom; and an operational dimension, since state awareness, organisational structure, data, and decision rights collectively determine whether a theoretically available option can actually be exercised in operation. Robust flexibility value arises from the interplay of all four dimensions.

2.3 Market and role logic

A flexibility market is a market mechanism through which economically addressable behaviour – rather than energy volume alone – is the primary object of exchange. The subject is the capacity to operate injection, demand, storage, or conversion differently from the baseline under defined conditions. This distinction separates pure energy delivery from valued responsiveness.

This responsiveness becomes visible across different market layers. The day-ahead and intraday markets form the horizontal reference space of electricity trading, organising price formation, schedule setting, and near-term position adjustment. Balancing energy constitutes a standardised flexibility market for system equilibrium: in the German and European context this encompasses FCR (Frequency Containment Reserve), aFRR (automatic Frequency Restoration Reserve), and mFRR (manual Frequency Restoration Reserve), each with distinct requirements for response speed, activation logic, and capacity obligations. Local flexibility markets address degrees of freedom whose value depends substantially on the location of activation. Vertical flexibility markets connect multiple layers of network, balancing group, aggregation, and platform logic within a shared activation and remuneration space.

For *marketing architecture*, the distinction between direct marketing and route-to-market is central. Direct marketing denotes the sale of electricity directly into the market or through a marketing partner. Route-to-market is the broader category, describing the full commercial pathway through which an asset or portfolio is embedded in market and contractual spaces. The distinction matters because the chosen marketing pathway – rather than the technical plant alone – determines which degrees of freedom remain accessible thereafter.

On the *contractual side*, PPAs, CPPAs, and CfDs are read in this compendium as structures for the distribution of degrees of freedom rather than as isolated contract categories. They stabilise revenues and, in doing so, redistribute quantity obligations, profile responsibility, and operational flexibility amongst the parties. This is precisely why they belong at the centre of the flexibility discussion rather than at its margins.

Four *role concepts* carry the argument on the institutional side. The balancing group is the central translation structure between physical behaviour and market-compatible settlement; the balance responsible party maintains this structure in equilibrium. The aggregator bundles distributed degrees of freedom into marketable volumes. The virtual power plant is the operational integration architecture that brings together forecasting, state management, dispatch, and market communication for distributed connected assets. The IPP describes the business model perspective of the independent power producer, a role that can expand substantially beyond pure generation in flexibility-capable portfolios.

2.4 Asset, operational and portfolio logic

An *asset* is the physical unit or functionally coherent plant assembly from which a degree of freedom arises. A portfolio, in the fuller sense, comes into being when multiple assets are brought into a shared decision and action space rather than simply aggregated. A hybrid portfolio extends this space by combining asset classes whose degree-of-freedom and state logics complement one another. Economic value does not arise from the individual asset alone. It arises where degrees of freedom are related to one another, prioritised, and governed under conditions of conflict. Value stacking and revenue stacking become relevant precisely here. Both terms describe the multi-dimensionality of flexible portfolios, and both are best understood as requiring rigorous internal logic rather than additive ambition. Where the same power, the same state-of-charge window, or the same load corridor is claimed by multiple markets simultaneously, portfolio management begins in earnest: principled prioritisation rather than a wishlist.

The operational control axis of such a portfolio runs through telemetry, observability, forecasting, scheduling, dispatch, and settlement, in that deliberate sequence. Telemetry creates data access. Observability converts that access into reliable state knowledge. Forecasting generates expectation spaces. Scheduling translates these into planned operating modes. Dispatch responds to reality. Settlement converts performance into auditable financial outcomes. Only through this complete chain does a degree of freedom become an operationally governable one.

Risk takes on a correspondingly structured role. Shape risk, imbalance risk, congestion risk, and degradation risk mark the conditions under which degrees of freedom retain or lose their value in operation. Hedging in this context is the deliberate structuring of

uncertainty across contracts, market positions, physical degrees of freedom, and portfolio architecture – a broader discipline than price protection alone. The operating model – the operational target picture of roles, decision rights, processes, systems, and escalation logic – is the organisational architecture through which degrees of freedom are kept operationally governable and organisationally coherent across the portfolio.

Part I – Asset & engineering layer

3. Technical degrees of freedom and asset logics

3.1 Technical degrees of freedom as the starting point

The technical assessment of flexibility begins with asset behaviour. A degree of freedom is a genuinely usable room within real boundaries, not an abstract potential. These boundaries arise from physics, topology, protection and safety logic, state variables, service life, measurability, and grid connection conditions. For this reason, treating flexibility as a blanket property of a technology class is insufficient. The same technology can carry very different economic value depending on configuration, control strategy, reserve management, operating regime, and connection context.

Seven dimensions govern the assessment: power, duration, response speed, state-dependence, locational relevance, controllability, and verifiability. An asset may be strong in one dimension and structurally constrained in another. For subsequent market and portfolio logic, the relevant question is therefore not whether an asset is 'flexible' in the abstract, but for which applications and under which boundary conditions its degree of freedom remains robustly available.

This distinction is operational rather than academic. A BESS responds quickly, precisely, and bidirectionally, whilst remaining sensitive to state and ageing. An electrolyser opens flexible demand without offering the mirror-image storage logic of a battery. A heat pump can shift load, but only within thermal and comfort constraints. A data centre can open energy-economically relevant load corridors, yet not every workload is temporally or spatially dispatchable. The value of a portfolio therefore lies in the structure, combinability, and operational governability of its degrees of freedom rather than in the count of its assets.

3.2 Photovoltaics

Photovoltaics is a resource-dependent generation technology of high scalability, low mechanical inertia, and considerable system impact. Its core technical contribution lies in cost-efficient power production; its principal constraint lies in the absence of free dispatch over the primary energy source. PV does not generate flexibility in the symmetric sense. It shifts the conditions under which flexibility becomes valuable.

The immediately available degrees of freedom lie in regulated output reduction, ramp limitation, reactive power provision through the inverter stage, voltage support, and a degree of profile influence through layout and topology choices. The distinction between system impact and own flexibility deserves emphasis. PV carries high system impact, since it substantially influences midday surpluses, local voltage and infeed situations, and price profiles. Its own flexibility remains limited in the absence of additional coupling.

Positive flexibility above the current resource level does not exist. The degree of freedom of a PV plant therefore arises less from the asset itself than from its embedding: the quality of coupling with storage or flexible loads, the inverter and grid interface, and an operating and marketing model that explicitly accounts for curtailment, surplus, and profile shaping. PV is an asymmetric asset in flexibility terms, and at the same time a highly relevant source of degrees of freedom within hybrid portfolios.

3.3 Wind

Wind energy follows a different technical logic from photovoltaics. The primary energy is equally resource-dependent, yet the resulting degree of freedom is shaped by different generation profiles, different temporal correlations with load and price, typically higher full-load hours, and a different network-side impact. Wind is its own degree-of-freedom type rather than a temporal variant of PV.

The technically immediate degrees of freedom lie in output reduction, de-loading, ramp management at turbine or farm level, reactive power provision, and voltage support. In appropriate configurations, wind can also contribute to system-support services. The central point remains, as with PV: wind is resource-dependent and therefore not freely symmetrically dispatchable. The dominant flexibility lies in the deliberate reduction and management of an available resource.

Its system relevance is often more strongly transmission- and congestion-related than that of PV, particularly where high wind capacity is spatially concentrated and must be transported over significant distances. This gives rise to substantial degrees of freedom in the redispatch, curtailment, and network management context. The value of wind lies therefore in its position within a broader network, market, and portfolio system as much as in the individual asset.

3.4 Battery Energy Storage System(s)

The BESS is, within this compendium, the technically most versatile flexibility resource. Its strength lies in the combination of fast response, bidirectional power, precise controllability, and the ability to decouple generation and consumption temporally. At the same time, a BESS is not a universal standard instrument but a complex integrated system whose value is determined by state management, integration quality, and ageing discipline.

A systems perspective is the appropriate starting point. A BESS comprises not only cells but an electrochemical core, module and rack structure, power conversion system, battery management system, thermal management, protection and fire safety logic, energy management system, and network and communication interfaces. The same nominal MW/MWh combination can carry

very different operational quality depending on PCS design, BMS strategy, temperature management, protection philosophy, and EMS logic. A substantial portion of storage value arises in the integration rather than in the chemistry alone.

Among cell chemistries for stationary applications, LFP and NMC currently dominate; LTO, sodium-ion, and flow systems remain relevant for specific use cases. These distinctions carry technical weight, since the chemistries differ in energy density, power density, thermal stability, cycle life, safety profile, temperature behaviour, and cost structure. LFP is frequently attractive in stationary applications for its robustness, safety, and cycle durability. NMC delivers higher energy densities but demands a different safety and operating regime. LTO offers outstanding power capability and cycle life, positioned differently in the cost and energy space. Flow systems decouple power and energy systematically and become interesting where duration matters more than peak power density.

The technical suitability of a BESS can be assessed robustly only when energy capacity, charge and discharge power, energy-to-power ratio, response time, efficiency, C-rate, depth of discharge, cycle life, state of charge, and state of health are held as distinct parameters. A high-power, short-duration system is a fundamentally different degree-of-freedom carrier from a longer-duration, moderate-power system. For balancing energy, intraday arbitrage, peak shaving, PPA shaping, or local network services, the same cell chemistry therefore implies very different suitability profiles.

The engineering precision of a BESS ultimately resides in its ageing logic. Degradation arises through time, temperature, cycling, charge and discharge rates, the SoC windows operated, and the dispatch regime. The operative boundary therefore runs between short-term marketable and long-term value-durably governable rather than between technically possible and technically impossible. A storage schedule that looks nominally attractive can destroy substantial value through degradation, reserve erosion, and opportunity cost. The BESS is a degree of freedom with its own running cost: its technical substance must itself be actively managed.

3.5 Hydrogen and electrolysis

Hydrogen is relevant in this compendium principally as the output of a conversion pathway rather than as an electrical storage medium in the narrow sense. The central plant is the electrolyser, which from a flexibility perspective functions above all as a controllable load. Its degree of freedom lies in deliberate power consumption under market, network, and process conditions rather than in symmetric electrical power output. Alkaline electrolysis, PEM, and SOEC are the dominant technology pathways, differing in dynamics, minimum load, ramp capability, temperature window, efficiency, material requirements, and integration logic. For the flexibility context, load dynamics are of primary importance. PEM systems are typically better suited to volatile power environments. Alkaline systems are robust and industrially established, though operating within different boundary conditions. SOEC systems open different temperature and process spaces.

The electrolyser alone describes the degree of freedom only partially. Its actual usability is substantially shaped by water treatment, compression, intermediate storage, offtake structure, safety requirements, minimum loads, and start-up and shutdown logic. For this reason, hydrogen is treated here as a sectoral freedom space with its own technical and infrastructural logic rather than as a simplified long-duration storage asset.

3.6 Further system-relevant flexibility resources

3.6.1 Hydro

Hydro is the classical reference technology for high-quality flexibility. Run-of-river, reservoir, and pumped storage plants differ substantially in technical character, yet share the central advantage of being able to couple electrical power with physical storage volume or inflow potential. Hydro is therefore typically capable of longer duration, greater reliability, and greater energy-side robustness than electrochemical storage, whilst being far more tightly bound to geography, topography, and available development space. For this compendium, hydro matters above all as the reference image of what the energy system has traditionally regarded as high-quality flexibility: power, duration, availability, and the capacity for multiple simultaneous service roles.

3.6.2 Heat pumps

The heat pump is, from a flexibility perspective, a thermally coupled load. Its degree of freedom arises from the fact that electrical power consumption and ultimate heat requirement need not coincide at every moment. Buildings, buffer storage, thermal mass, and comfort bands open a time corridor within which load can be shifted. Technically central are thermal storage capacity, building thermal inertia, ambient temperature dependence, flow and storage temperatures, comfort limits, and controllability. Heat pumps are therefore highly relevant thermally buffered loads with substantial potential for load shifting, network-oriented application, and aggregation, operating within defined temporal and temperature envelopes rather than as universal short-term flexibility instruments.

3.6.3 EV charging infrastructure

EV charging infrastructure is technically a controllable load. Its degree of freedom arises from the difference between the energy quantity required and the time window within which that quantity must be delivered. Technically decisive are charging power, charging time windows, user behaviour, vehicle state, communication standards, site structure, and grid connection. The strength of this asset class lies in its aggregability and digital controllability rather than in the individual charging point. Many individually modest degrees of freedom can, through aggregation, constitute a network- and market-relevant action space.

3.6.4 Data centres

Data centres are critical high-density loads with demanding availability requirements and growing energy-economic significance. Their economically relevant degrees of freedom lie particularly where they can couple generation, storage, cooling, load

management, and backup structures within their own operational control space behind the meter – that is, on the consumption side behind the site's grid connection point. The degree of freedom therefore resides in the coupling of energy infrastructure, cooling, backup architecture, load management, and digital control rather than in unconstrained variability of the IT core load. Five areas typically carry energy-economic relevance.

- 1) UPS, BESS, and backup structures that provide not only resilience but, under defined conditions, fast response capability, short-term buffering, and controlled grid decoupling.
- 2) Local generation – PV in particular – which, when coupled with BESS and load control, extends the behind-the-meter action space.
- 3) Cooling architecture, where thermal inertia and admissible temperature windows can open load corridors of energy-economic relevance.
- 4) Energy-aware compute control: a control logic that relates price, carbon, weather, renewable, and grid signals to state awareness, storage management, and operating modes.
- 5) Compute or workload scheduling, applicable to those load components that are neither strictly latency-critical nor SLA-constrained.

The energy-economic value of data centres deserves neither underestimation nor overstatement. Resilience and availability requirements take precedence over revenue optimisation. Not every UPS or BESS configuration constitutes marketable flexibility. Not every computing load is temporally or spatially shiftable. Thermal upgrading through waste heat recovery remains dependent on temperature level, distance, heat sink, hydraulics, and seasonal coincidence. Within these boundaries, however, data centres combine electrical load, storage, cooling, data, and control logic in a way that increasingly makes them active, steerable energy nodes within the broader system.

3.6.5 Energy-intensive industrial processes

Energy-intensive industry forms a flexibility class of its own, distinct from the household-scale and infrastructure loads considered above. Its degree of freedom resides in process load that can, within limits, be reduced, raised, or shifted in time. Three operating modes structure this space. Load reduction lowers drawn power below the reference operating point and provides positive flexibility, usually at the cost of foregone or deferred production. Load increase raises consumption and provides negative flexibility, available only where spare production capacity and downstream absorption exist. Load shifting combines the two, so that curtailed output is recovered within a short window and net production is preserved. The governing engineering parameters are therefore the recovery obligation, the buffering available upstream and downstream, and the admissible deviation of process parameters, alongside power and duration.

The processes that carry this flexibility differ markedly by sector. Electric arc furnaces in steelmaking can interrupt for short intervals and offer substantial positive potential; aluminium electrolysis tolerates a load variation in the region of a quarter of nominal power within cell-stability limits; cement raw and finishing mills decouple from the kiln through silo buffering; glass tanks open flexibility through hybrid electric-and-gas firing; and chlor-alkali electrolysis operates across a wide load band where intermediate product storage permits. Alongside these branch-specific processes sit cross-cutting technologies – process heat, cooling, compressed air, pumps, and hybrid heat supply – whose flexibility is accessible to a broader range of sites, including smaller ones.

Industrial flexibility also makes a general point precise. A degree of freedom is rarely a single figure; it is a cascade of progressively narrower potentials. A theoretical potential is reduced by safety and plant constraints to a technical potential; within that, only the economically worthwhile and the practically admissible portions are realisable; and only the share actually contracted and exercised in a given period is finally available to a market. Reading flexibility at the appropriate point in this cascade guards against both the overstatement of headline potential and the neglect of genuinely usable room.

3.7 Multi-use capability as an engineering design requirement

The asset logics set out above share a consequence that belongs in the engineering phase rather than in later optimisation. Where the same physical resource is to serve own-use, market, and network-oriented purposes in turn, multi-use capability must be designed in from the outset. It is seldom available as a retrofit to an asset conceived for a single application. Grid connection, protection and metering concept, control and communication interfaces, and the operating envelope each determine whether a degree of freedom can subsequently be activated compatibly across several logics.

Several engineering properties become load-bearing here. Activation compatibility describes whether an asset can respond to different signals without mutual interference. The technical availability window defines the periods in which a degree of freedom is genuinely free rather than already committed by state or reservation. The network-constrained operating envelope captures the local limits within which an activation remains admissible. Rebound risk recognises that a reduction frequently implies a later recovery, which can itself create a new peak if it is left unmanaged. Prequalification readiness determines whether an asset can be admitted to a given product at all, and metering and baseline capability decide whether its contribution can later be measured, validated, and settled. An asset strong in nominal power but weak in these properties carries a correspondingly constrained flexibility value.

Network-oriented activation in particular presupposes controllability, state knowledge, communication capability, and secure intervention rights at the connection point. Designing these in early is modest in cost relative to retrofitting them later, and it is what allows a single resource to move between own-use optimisation, market participation, and network service without renegotiating its physical and contractual basis on each occasion. Multi-use is, in this sense, an engineering decision before it becomes a commercial one.

4. Technical suitability, hybridisation, and controllability

4.1 Technical suitability by market application

An asset's flexibility is always relative to a specific application. Technical suitability must therefore be read in relation to time horizon, product logic, locational context, and operating objective.

For the day-ahead market, forecast and schedule capability are decisive. PV and wind shape volume and price structure; storage and flexible loads complement this layer through temporal shifting and profile management. In the intraday market, the value of short-term response capability rises; BESS is particularly strong here, whilst electrolyzers, heat pumps, and aggregated charging infrastructure can each become relevant depending on the operating regime. Balancing energy demands response speed, availability, prequalification capability, and reliable activation – applicable above all to FCR, aFRR, and mFRR, which impose distinct requirements on speed, reservation, and activation logic. Storage assets are naturally well positioned; other asset classes can qualify within their own technical envelopes.

For congestion management and local flexibility markets, locational specificity is the governing criterion. PV and wind can contribute negative flexibility; BESS provides local buffering; electrolyzers offer flexible load uptake; heat pumps and charging infrastructure offer shiftable demand; and data centres can expose defined load corridors. For capacity and adequacy frameworks, duration, reliability, and reservation capability are the governing dimensions. Technical suitability is therefore always a function of asset character, market product, and state regime rather than an intrinsic property.

Asset	Power	Duration	Response	Market fit	Constraint
PV	Medium	Resource-dependent	Fast	DA, local	Resource availability
Wind	High	Resource-dependent	Medium	DA, ID, local	Resource availability, network
BESS	High	Short	Very fast	ID, balancing services, local	SoC, degradation
Electrolysis, H ₂	Medium	Variable, medium to long	Medium	ID, local	Minimum load, process constraints
Heat Pump	Low	Medium	Medium	Local, DSM	Comfort, thermal constraints
Hydro	High	Long	Very fast	DA, ID, balancing services, local	Availability, storage level
EV	Aggregated	Short to medium	Fast	Local, DSM	Charging window
Data Centre	High	Variable	Medium	Local, behind-the-meter	SLA, resilience

Table 1: Technical flexibility profiles of key asset classes – power, duration, and market characteristics compared

4.2 Hybridisation as a co-optimisable design space

Hybridisation is a design space in which degrees of freedom are deliberately combined to create new action opportunities, rather than a simple technical co-location of multiple technologies. The value lies in the quality of coupling rather than in the number of components.

In PV-and-BESS and wind-and-BESS configurations, the primary gains lie in better utilisation of the grid connection, absorption of surpluses, curtailment reduction, profile smoothing, and additional response capability. In hydrogen-and-BESS configurations, the logic shifts: value arises from combining fast electrical buffering with controllable demand, longer-duration conversion, and sectoral connectivity. Where flexible loads such as heat pumps, EV charging infrastructure, or data centres are added, the design space extends further into demand-side co-optimisation.

Hybridisation is therefore an architectural question from the outset. Grid connection, power electronics, SoC management, charge and discharge priority, metering concept, dispatch logic, and marketing pathway must be conceived together. The additional value arises from a coupling that brings degrees of freedom into a shared, operationally governable decision space – not from additional technology as such.

4.3 Data, telemetry, and controllability as technical preconditions

Technical suitability does not end at the hardware boundary. A degree of freedom remains below its economic and operational potential where it cannot be observed, forecast, and commanded. Telemetry delivers state and operating data. Observability converts that data into reliable state knowledge. Forecasting generates anticipation. Dispatch translates decisions into operational action.

The real distinction runs between an asset whose degree of freedom is technically present but operationally inaccessible, and one whose behaviour can be translated into a robust market, contractual, and operational framework. Digital capability in the flexibility context is not a parallel world but the connective tissue between engineering, market participation, and portfolio management. Without reliable data, precise control is unavailable. Without precise control, market-compatible and contractually robust activation is unavailable. Digitality is therefore an enabling condition for the realisation of physical optionality rather than an optional enhancement.

Part II – Market & contract layer

5. Market architecture and flexibility spaces

5.1 Market logic and layer differentiation

Flexibility does not become economically visible within a single market but across several differently structured coordination spaces. These differ in time horizon, product logic, degree of standardisation, activation regime, locational relevance, liquidity, and regulatory embedding. Describing flexibility as an additional revenue stream or as a sublayer of the wholesale market

therefore misses the point. Its value arises where a degree of freedom remains accessible across multiple application and revenue spaces without allowing technical, balancing, or contractual boundaries to be overlooked in the process.

Three layers can be distinguished at the core. The *wholesale markets* – day-ahead and intraday above all – organise price formation, energy allocation, and near-term position adjustment within standardised, liquid market environments. The *balancing energy markets* procure and activate standardised short-term flexibility to secure system equilibrium; here the emphasis falls on availability, response speed, reliable activation, and product capability rather than on energy volumes as such. Local and vertical *flexibility markets* then address those degrees of freedom whose value is bound to location within the network, direction of activation, and the coordination between network, balancing group, aggregation, and platform logic.

These layers are most usefully read as complementary coordination strata of the same energy-economic reality rather than as competing market models. Day-ahead and intraday markets deliver robust large-area price and schedule signals. Balancing energy provides standardised system flexibility. Local and vertical flexibility markets serve both as an institutional complement and as an attempt to bring better local information earlier and more economically into the dispatch and activation logic of the system. Local markets demonstrate with clarity that product design, prequalification, activation logic, data access, settlement, and the allocation of roles and responsibilities are preconditions for functional market integration rather than downstream detail.

5.2 Direct marketing and route-to-market

The economic translation of an asset begins with its route-to-market: the full commercial pathway through which an asset or portfolio is embedded in market and contractual spaces. Direct marketing is one concrete expression of this pathway, connecting generation or flexibility behaviour with forecasting, schedule management, balancing group assignment, redispatch processes, and financial settlement.

The distinction matters because the same technical plant can carry very different economic degrees of freedom under different routes-to-market. A technically attractive asset managed through a marketing pathway that structurally constrains profile freedom, storage dispatch, or local market participation will realise only a portion of its available value. Conversely, a well-designed marketing architecture can deliberately conserve degrees of freedom and thereby keep later optimisation genuinely open.

This effect is frequently underestimated in early project phases, when route-to-market appears to be a pricing or distribution question. Its significance becomes clear later, when it proves to have been a primary determinant of portfolio mobility. For PV-and-BESS, wind-and-BESS, and hydrogen-and-BESS portfolios, the relevant question is therefore which marketing pathway preserves action space, which risks it absorbs, and which operating model it presupposes – alongside the familiar questions of bankability and marketability.

5.3 Power purchase agreements and corporate power purchase agreements

PPAs and CPPAs function in this compendium as architectures for the distribution of degrees of freedom rather than as simple price contracts. They determine how revenues are stabilised and, simultaneously, how quantity, profile, and market risks are distributed amongst the parties, and how much flexibility remains within the portfolio.

A contract oriented closely to actual generation typically preserves greater profile proximity to physical reality whilst leaving more market volatility within the downstream marketing pathway. A more shaped or structured delivery profile provides greater usage stability from the offtaker's perspective, whilst increasing the internal demands on storage management, load control, balancing group logic, and portfolio optimisation. This is precisely where PPA structures reveal themselves as instruments that reshape technical and operational degrees of freedom rather than simply providing commercial security.

The logic sharpens in hybrid portfolios. A PV-and-BESS portfolio can support contract profiles differently from a pure PV plant. A wind-and-BESS portfolio shifts the profile and timing logic. A hydrogen-and-BESS portfolio brings together power procurement, conversion, storage capability, and potentially molecular offtake within a different degree-of-freedom structure. The productive question is therefore never whether a PPA is appropriate in principle, but which degrees of freedom a specific contract preserves, which it binds, and at which point it displaces additional complexity into the portfolio management function.

5.4 Contracts-for-Difference and related price and revenue structures

The Contract for Difference ("CfD") is a price-stabilisation structure whose primary economic function lies in smoothing price risk and thereby improving planning certainty and financeability. For the present compendium, this is the starting point rather than the conclusion. The more consequential question is the extent to which such a structure keeps additional degrees of freedom open or, conversely, narrows the portfolio's action space.

Every price-hedging structure modifies optionality. Where revenue is tied tightly to reference quantities or reference prices, the economic significance of additional market or optimisation freedom diminishes. Where temporal or quantity-related degrees of freedom remain open, price stabilisation can be compatible with storage management, intraday optimisation, and local flexibility participation. The essential distinction therefore runs between rigid and mobile risk structures rather than between hedged and unhedged positions.

In this sense, the CfD belongs to the same analytical family as the PPA, direct marketing, and hedging. It is a structure through which price risk is smoothed whilst simultaneously determining how much operational flexibility the portfolio retains thereafter. The current debate also supports reading CfDs as investment and risk structures for broader decarbonisation pathways – beyond mature electricity generation alone – in which the reference logic, market integration, and degree-of-freedom effects require careful calibration. Table 2 sets out this interplay across the principal market and contract mechanisms.

Mechanism	Pricing logic	Profile linkage	Effect on flexibility potential	Typical conflict
DA	Zonal	Medium	Limited and planned	Forecast error
ID	Short-term	Low	High	Liquidity, timing
Balancing Services	Standardised	Reserve commitment	High, but capacity is reserved	SoC, availability constraints
Local Flexibility Market	Locational	Situational	High, but location-bound	Balancing group, ToE
PPA	Contractual	High, variable	Depends on contractual structure	Shape, volumes
CfD	Reference price	Indirect	Price is stabilised; flexibility potential depends on design	Dispatch incentive

Table 2: How key market and contract mechanisms shape price logic, profile commitment, and retained operational degrees of freedom

5.5 The balancing group, the balance responsible party, and the market side of physical effects

The balancing group is the central translation structure between physical behaviour and market-compatible order. The balance responsible party carries responsibility for keeping that structure in equilibrium. For flexibility, this is a decisive interface between market and physics rather than an administrative afterthought.

Every material activation changes not only asset behaviour but, in the great majority of cases, the bilaterally expected position within the balancing group. A load is shifted, a storage asset charges rather than discharges, generation is curtailed, or an electrolyser absorbs additional power within a specific time window. These changes propagate into schedules, imbalance costs, marketing logic, and potentially contractual obligations. Balancing group logic is therefore constitutive for marketable flexibility.

Many flexibility models remain too abstract at precisely this point. They describe technical degrees of freedom accurately whilst underestimating their effect on balancing and settlement. Multiple uses, aggregation, or activation conflicts then reveal that the same flexibility was committed in parallel or evaluated inconsistently across different spheres. Balancing consequences must be part of the design from the outset for flexibility to be marketed with confidence.

5.6 Aggregator, virtual power plant, and independent power producer

The aggregator bundles distributed degrees of freedom into marketable volumes. Its contribution lies not only in quantity aggregation but in translating heterogeneous technical potentials into product capability, planability, activatability, and settlement. Where individual assets are too small, too volatile, or too organisationally fragmented, aggregation becomes the precondition for market participation rather than an optional enhancement.

The virtual power plant is the operational architecture within which this bundling is brought together. It connects asset integration, forecasting, state management, schedules, dispatch, and market communication. The VPP is an integration and control structure rather than a market or an ownership role.

The IPP, in turn, describes the business model perspective of the independent power producer. In flexibility-capable portfolios, this role can extend substantially beyond pure generation to encompass storage, loads, contract logic, optimisation, and platform integration.

The conceptual separation remains important. Aggregation is a bundling function; the VPP is an operational integration architecture; the IPP is a business model and ownership perspective. Only when these layers are cleanly distinguished can one assess where market logic ends and where the operational architecture of the portfolio begins.

6. Balancing energy, local markets, and commercial conflict logic

6.1 Balancing energy as a standardised flexibility market

Balancing energy is the standardised market for short-term activatable system flexibility. Its subject is not simply energy but the capacity to deliver defined power reliably within clear temporal and product logics. In operational market practice this manifest principally in FCR, aFRR, and mFRR, each imposing distinct requirements on response speed, activation logic, and reservation obligations. Reservation, prequalification, response time, activatability, and availability therefore occupy a more central position here than in conventional wholesale transactions.

This is where its operational sharpness lies. A portfolio addressing balancing energy does not merely open a potential revenue space; it commits degrees of freedom. A storage asset reserving capacity for balancing energy sets aside power and SoC window. A flexible load held available for activation cannot simultaneously deploy the same room freely for arbitrage or local congestion absorption. Balancing energy is therefore a structural intervention in portfolio availability as much as a market.

Two reasons make this particularly important for the compendium. Balancing energy is itself a flexibility market and should not be separated from the flexibility discussion. Its standardisation also makes especially visible how strongly economic relevance is bound to technical discipline, product capability, and operational governability.

6.2 Local and vertical flexibility markets

Local and vertical flexibility markets address those degrees of freedom whose value is bound to the location of activation, the concrete network state, and the coordination of multiple system levels. Where wholesale markets organise large-area energy allocation and balancing energy addresses standardised system stability, these markets access the values arising from network node, direction, bottleneck, and institutional coupling.

Their distinctive character lies in mapping physical reality with greater resolution, whilst typically being less standardised and often less liquid than the wholesale layer. They are neither a countermodel to wholesale trading nor a narrow speciality niche,

but an expression of a system reality in which zonal prices and standardised products do not transmit all relevant signals with equal precision.

For a portfolio, this means that local and vertical markets open an additional prioritisation dimension rather than a frictionless additional revenue layer. A degree of freedom that is highly valuable locally may carry low opportunity cost in wholesale trading – or the reverse. Precisely in this differential lies the market intelligence of flexible portfolios.

The regulatory substrate of these markets is itself differentiated. Instruments for activating flexibility in the network context fall into three families. Market-based procurement allows the network operator to contract capacity-restriction or redispatch-type products from connected parties or service providers. Non-market-based intervention is retained as a regulated last resort, used where competition is insufficient or congestion so predictable that a market would invite gaming. Network-tariff instruments shape behaviour through price signals rather than direct contracting. Where the market-based products are activated through dedicated platforms, the relevant point for portfolio design is the contractual obligation they create rather than the platform that carries them.

Continental practice adds further instruments of direct relevance to flexible portfolios. Time- or load-variable network charges introduce a price signal that rewards network-compatible behaviour, where static charges reward only uniform or minimal withdrawal. Differentiated network-usage products distinguish firm from conditional network use, allowing a user able to curtail or shift to accept a conditional connection at a lower charge. Contractable flexibility products allow the network operator to remunerate connected parties for grid-serving deployment they would not otherwise choose. Two structural obstacles recur: state-induced price components levied per unit of energy can distort the economics of conversion and sector-coupling loads, and the asymmetric treatment of capital and operating expenditure within incentive regulation can favour conventional reinforcement over operationally provided flexibility. For the portfolio, these instruments determine how much of a technically available degree of freedom is also a contractually and economically accessible one.

6.3 Incentivisation, participation, and liquidity

Technically available flexibility becomes market-effective only where its provision appears economically and organisationally worthwhile. Decisive here are not only prices but entry barriers, product logic, minimum size thresholds, data requirements, settlement security, and regulatory compatibility. In practice, flexibility fails less frequently through physics than through market architecture.

The quality of a flexibility market is therefore not determined by its formal existence alone. A market can be theoretically well constructed yet practically weak where participation frictions are high, products are too narrowly defined, revenue prospects are too uncertain, or attribution logics are too complex. In local markets this risk is particularly pronounced, since physical relevance and economic depth can diverge sharply. Local flexibility markets demonstrate in addition that prequalification, activation logic, data access, settlement, and the assignment of roles and responsibilities are preconditions for robust market integration rather than downstream administrative questions.

Liquidity is accordingly a structural condition rather than a comfort variable. Without sufficient participants, volumes, and product compatibility, a market remains illiquid and fragile. This gives aggregation, standardisation, and platform capability their strategic significance: they do not increase the physical flexibility available but raise the proportion of technical degrees of freedom that can be translated into actual market participation.

The economic function of these markets extends beyond additional revenue creation to potentially more efficient utilisation of existing infrastructure and available energy. Where degrees of freedom can be activated earlier, more locally, and more precisely, the need for downstream correction mechanisms diminishes. This can reduce redispatch and curtailment requirements, improve utilisation of network and generation capacity, and simultaneously enhance the integration of renewable infeed. The environmental benefit arises as a systemic co-effect of more efficient and locally better-informed coordination rather than as an independent objective.

6.4 Energy-as-a-Service and platform ecosystems

Energy-as-a-Service („EaaS“) denotes, within this compendium, a shift in the service perspective rather than a marketing label. Energy and flexibility services are offered not as isolated products but as integrated, data- and platform-supported solution architectures. This is particularly relevant for flexibility because degrees of freedom rarely appear in isolation in practice. Their economic relevance arises frequently from the interplay of forecasting, dispatch, balancing, contract logic, and customer or site context rather than from any single technical capability.

A platform ecosystem provides the technical and institutional infrastructure for this interplay. Platforms reduce friction in registration, qualification, data exchange, activation, monitoring, and settlement. They create no new physical flexibility. They do, however, raise the proportion of technical degrees of freedom that becomes operationally and commercially accessible. In local and vertical markets particularly, this matters considerably, since market participation otherwise encounters organisational complexity that can be prohibitive.

6.5 Value stacking and revenue stacking

Value stacking and revenue stacking describe the multi-dimensionality of flexible portfolios and are best understood as calling for rigorous internal logic rather than additive ambition. An asset can make multiple system contributions and address multiple revenue spaces. The simultaneous realisation of all these contributions without conflict is, however, a different matter.

The decisive point is scarcity. Power, energy content, SoC window, load corridor, and activation capability are each finite. A storage asset reserving capacity for balancing energy cannot simultaneously deploy the same degree of freedom in full for arbitrage, local

network services, and PPA shaping. An electrolyser intended to absorb negative prices or network congestion is not simultaneously available without limit as a production optimiser. The distinction between robust stacking and opportunistic overcommitment lies in the discipline with which degrees of freedom are allocated across multiple value contributions.

Evidence from studies of multi-use storage and flexibility operation supports reading this discipline as a source of value rather than a constraint. Network-driven adjustments to an otherwise commercially optimised schedule are infrequent: in detailed case work they arise in only a few per cent of annual operating hours and remain below a tenth of operating time even in congested networks. The economic cost of operating network-compatibly rather than purely commercially is correspondingly small, on the order of a couple of per cent of the operator's result, while the network benefit, in the majority of cases examined, is cheaper than the conventional reinforcement it defers. The implication for portfolio design follows directly. Multi-use does not require that every application be served at once; applications can alternate, and a disciplined priority logic that yields to the network only when it must can preserve most of the commercial value while opening a network-serving contribution that defers capital expenditure at system level.

6.6 Conflict logics between revenue spaces

Once a portfolio addresses multiple markets and contractual spaces, conflicts arise. These are features of portfolio reality rather than exceptions, and their management is the substance of portfolio intelligence.

- The first conflict type is physical. A degree of freedom cannot be deployed simultaneously in contradictory directions. The same power, the same SoC window, or the same load corridor can be committed only once.
- The second conflict type is contractual. PPA, direct marketing, CfD, or supply structures can bind degrees of freedom that would be attractive from the perspective of other markets. Contractual stability and operational flexibility stand not in an either-or relationship but in a structured tension that must be managed consciously.
- The third conflict type is institutional. Different market roles, balancing group logics, activation rights, network requirements, and settlement regimes can render a technically sensible deployment economically or administratively unattractive. Where the energy and flexibility supply chains diverge, these conflicts become particularly prominent.

The quality of a market and contract model reveals itself in the consistency with which degrees of freedom are assigned, prioritised, and delimited under conflict conditions rather than in the number of potential revenue streams it nominally addresses.

6.7 Commercial conclusions

The market and contract layer shifts the question from whether an asset is technically flexible to how that flexibility can be economically preserved, activated, and settled. Wholesale markets deliver price and schedule signals. Balancing energy constitutes the standardised flexibility market for short-term system stability. Local and vertical markets access location- and network-specific values. Direct marketing, route-to-market, PPA, CPPA, and CfD structures redistribute risk and degrees of freedom. Aggregators, VPP architectures, platform ecosystems, and EaaS models condense technical potentials into marketable structure.

This sets up the question to follow: how are these degrees of freedom hedged against risk within the portfolio, prioritised in real time, and durably implemented across ETRM, governance, dispatch, and settlement?

Part III – Operating & portfolio layer

7. Portfolio management, risk, and optimisation

7.1 From asset to portfolio

The transition from individual asset to portfolio is a change in control logic rather than a simple increase in scale. An asset can be dispatched within defined technical boundaries. A portfolio must additionally synchronise degrees of freedom, weigh risks against one another, coordinate market and contractual spaces, and set priorities under uncertainty. This is precisely where the logic of isolated plant operation ends, and the logic of an integrated action space begins.

A portfolio comes into being in the fuller sense not when multiple assets appear in the same overview, but when technical degrees of freedom, market positions, contractual commitments, balancing groups, state variables, and risk limits are brought into a shared decision architecture. The assessment criterion shifts accordingly. A degree of freedom is no longer evaluated by its isolated individual value but by its marginal contribution relative to alternative uses and by its effect on the robustness of the portfolio.

A storage asset is therefore valuable not because it can charge and discharge, but where its power, energy content, SoC window, ageing costs, market access, and contractual context are aligned such that its deployment makes the highest risk-adjusted contribution to the overall portfolio. The same applies to flexible loads, curtailment space, electrolysis windows, local market options, and shaped delivery structures. Portfolio management begins where the relationship between alternatives is optimised rather than each individual activity in isolation.

7.2 Portfolio risks and their structure

Every additional degree of freedom increases the density of risks alongside the breadth of the action space. These risks are rarely isolated. In practice they overlap, reinforce one another, and shift in significance depending on market and operating conditions. A robust operating model treats them as an interconnected system rather than as a checklist.

Price risk concerns the uncertainty of future market prices and price spreads across day-ahead, intraday, balancing energy, and where applicable local markets. For flexible portfolios, the structure of price relations across time, market, and activation probability matters

more than any single price level. Volume risk concerns the uncertainty of available generation, load, or flexibility – resource-driven for PV and wind, state- and availability-related for storage, process- or usage-dependent for flexible loads.

Shape risk describes the uncertainty of whether the temporal profile of an asset or portfolio matches the price windows, delivery profiles, or obligations within which economic value arises. Its weight grows considerably with increasing contract and hybrid complexity. Imbalance risk concerns deviations between planned and actual behaviour, acting directly on balancing group management, imbalance costs, and operational response requirements. Congestion risk describes the risk that network bottlenecks impair planned operating modes or marketing logics – whilst also, in some configurations, opening additional local opportunities. Degradation risk affects BESS above all, though in different forms it applies to further systems whose technical value erodes under unfavourable dispatch regimes.

Liquidity and collateral risks arise particularly in active market participation and under margining requirements, whilst operational risks stem from data errors, communication failures, model deficiencies, mis-activations, and unclear responsibilities. A portfolio does not become unmanageable through this complexity. It does require a risk logic that addresses technical, market, contractual, and procedural risks in their coupling rather than in artificial separation.

7.3 Hedging as a structuring function

Hedging in a flexibility-capable portfolio is a structuring function rather than a downstream trading discipline. Its purpose lies in the deliberate ordering of uncertainty across financial counter-positions, physical degrees of freedom, contract structures, and reserve allocations rather than in price smoothing alone.

Simple price hedging can reduce revenue volatility. For flexible portfolios, this is insufficient as a complete risk framework. A storage asset can smooth profile risks as well as exploit price spreads. A PPA can stabilise revenues whilst simultaneously binding operational degrees of freedom. An electrolyser can absorb negative price windows, influencing both market and network risk in the process. A load control logic can smooth peak loads, reduce balancing deviations, and create local activation capability. Hedging is therefore one of the load-bearing functions of portfolio management rather than an addition to it.

The practical core lies in the reserve and allocation question. Which degrees of freedom are deployed for revenue generation, which for resilience, which for contractual stability, which for imbalance reduction, which for local network effect? Only at this point does it become clear that hedging in flexible portfolios concerns the deliberate reservation and structuring of physical and contractual freedom spaces as much as it concerns financial positions.

7.4 The portfolio logic of hybrid structures

Hybrid portfolios derive their value from a change in correlations rather than from technical diversity as such. PV, wind, BESS, electrolysis, heat pumps, charging infrastructure, and data centres respond differently to weather, prices, congestion, activation signals, and contractual requirements. Portfolio value arises where these differences are brought together deliberately and governed operationally rather than simply coexisting.

A PV-and-BESS portfolio can absorb midday surpluses, reduce curtailment, exploit intraday windows, and stabilise shaped delivery. A wind-and-BESS portfolio opens different time spaces and can buffer transport and profile risks differently. A hydrogen-and-BESS portfolio combines fast electrical buffering with controllable demand and longer-duration conversion. Flexible loads such as heat pumps, EV charging infrastructure, and data centres extend this logic to the demand side, making portfolio optimisation a system management question as much as a generation question.

The added value of hybrid structures is not automatic, however. Additional asset classes initially enlarge the decision space rather than improving its quality. Whether economic maturity follows depends on three conditions: are the degrees of freedom technically compatible? Can they be embedded in consistent market and contract structures? And is their shared operational governability robust? Where all three conditions hold, hybridity becomes a genuine portfolio advantage.

7.5 Optimisation logic: hierarchy over simultaneity

A portfolio cannot fully exploit all market opportunities simultaneously. The same power, the same SoC window, or the same load corridor can be committed only once. A robust optimisation logic therefore begins with hierarchy rather than simultaneity. The first level is the structural action space. Here, technical safety boundaries, contractual commitments, regulatory conditions, balancing group logic, minimum reserves, and risk limits are established. This level defines what is permissible. The second level is the planning allocation, where day-ahead positions, contract profiles, SoC targets, reservations, and base allocations are set. The third level is operational re-optimisation, encompassing intraday adjustment, local activation, balancing energy deployment, state updating, and the management of forecast deviations. The fourth level is feedback, where deviations, opportunity losses, limit breaches, state drift, and KPI signals are analysed and translated into revised decision rules. This hierarchy is an expression of operational maturity rather than a theoretical ordering. A portfolio is robustly positioned when it addresses many markets and can prioritise amongst them on principled grounds.

7.6 Forecasting, scheduling, and real-time management

A flexibility-capable portfolio depends on the capacity to anticipate the future and to respond to changed conditions quickly and under control. Forecasting, scheduling, and real-time management form the operational core axis for this purpose.

Forecasting generates expectation spaces for generation, load, prices, states, congestion, and activation probabilities. Scheduling translates these expectation spaces into schedules, reserve holdings, delivery profiles, SoC targets, and load windows. Real-time management responds to reality: weather deviation, network signal, activation call, contract deviation, or system disturbance.

The quality of this control axis lies in its continuity rather than in the existence of individual tools. Sound forecasting without operational connectivity remains weak in effect. A strong dispatch function without a reliable forecast basis becomes reactive and costly. A storage asset without disciplined SoC management loses market value or reserve capability. A data centre without energy-aware control logic remains largely passive in energy-economic terms despite its digital infrastructure. Operational governability arises where data, state knowledge, planning, and response converge in a consistent cycle.

8. Operating model, ETRM, and governance

8.1 Energy trading and risk management as operational backbone

ETRM forms the operational backbone of a flexibility-capable portfolio. It brings together market positions, price references, counterparties, schedules, balancing logic, limits, settlement, and reporting within a shared management and transparency space. In simple trading environments, ETRM can be understood as a position and risk system. In flexible portfolios its significance grows considerably, since technical states, market opportunities, contractual commitments, and operational reservations must be coordinated simultaneously.

A storage asset in the ETRM context is simultaneously a position with opportunity value, marginal cost, risk function, and settlement consequence. The same applies to flexible loads, shaped PPA obligations, balancing energy reservations, and local activation windows. Without a system that maps these layers consistently, the portfolio remains either manual, opaque, or unscalable. ETRM is therefore the operational structure within which technical reality and commercial order converge rather than a trading tool in the narrow sense.

8.2 The trading lifecycle of flexibility

The classical trading lifecycle applies to flexibility with adaptation rather than unchanged. At its centre lies not simply a tradeable energy product but a state-dependent degree of freedom whose value arises from technical availability, market access, activation capability, contract logic, balancing group effect, and downstream verifiability.

Pre-trade therefore begins with qualification. The decisive questions are whether a degree of freedom is technically available in the relevant time window, commercially permissible, operationally reservable, and free of conflict across its various potential uses. Trade encompasses bid submission, market or platform access, and the reliable transfer into an activatable dispatch state. Post-trade extends beyond clearing and settlement to encompass measurement, validation, balancing group reconciliation, where applicable transfer of energy, and an auditable performance and revenue attribution.

For flexible portfolios this means that trading, dispatch, and operating model form a continuous decision chain rather than separate layers. Optionality becomes economically effective only where it remains physically deliverable, contractually consistent, and operationally verifiable. Automated bidding and optimisation logics are correspondingly relevant, yet remain bound throughout to governance frameworks, limits, and auditable decision rules.

8.3 Portfolio governance, limits, and decision rights

The broader the action space, the greater the need for clear decision rights. Informal coordination is often adequate in simple asset contexts. In flexibility-capable portfolios with multiple markets, products, contractual commitments, and activation logics, it becomes a source of error. Governance here denotes the totality of roles, rules, limits, escalation pathways, override rights, and control mechanisms through which degrees of freedom are managed in an orderly fashion.

This encompasses responsibilities across trading, asset management, market operations, dispatch, risk, finance, and IT; technical and commercial limits; reserve and SoC specifications; rules for resolving conflicts between revenue spaces; documentation obligations; and intervention and escalation processes. Governance is an operational necessity rather than bureaucratic framing. Trading optimises degrees of freedom differently from asset management. Operations assesses risks differently from commercial functions. A balancing energy deployment can conflict with intraday logic; a local activation can conflict with shaped delivery; a BESS deployment can conflict with degradation protection. Without clear governance, degrees of freedom become more contentious rather than more accessible.

8.4 Measurement, validation, and settlement

Flexibility becomes economically robust where its delivery is verifiable, its effect attributable, and its remuneration consistently settleable. Measurement, validation, and settlement form the operational foundation for this.

Measurement captures actual behaviour. Validation examines whether an activation was delivered in the expected scope, under the applicable rules, and without conflict with other commitments. Settlement translates this validated performance into financial and balancing settlement. Particularly in the context of aggregation, baseline methodology, transfer of energy, balancing energy, local markets, and hybrid portfolios, this layer determines whether technical possibility becomes institutionally robust flexibility.

For the operating model this is consequential. A degree of freedom that is technically present but cannot be cleanly measured, validated, and settled remains commercially fragile. A product without robust settlement loses market acceptance. In a mature portfolio, this layer is a core function rather than a back-office activity.

The verifiability on which this layer depends rests in turn on a digital, data, and cyber foundation. A degree of freedom is auditable only where telemetry, observability, and a consistent schedule source make its state and its activation visible; controllability evidence and audit-ready dispatch are properties of the data and control architecture before they are properties of the contract. Operational-technology security is therefore an operating condition for flexibility rather than a separate compliance matter. Where grid-serving activation depends on the remote control of assets in critical infrastructure, the

integrity of the IT/OT interface, the ability to detect and contain incidents, and the capacity to recover control after disruption become preconditions for cyber-resilient activation and, increasingly, for bankability and regulatory admissibility.

Where data-driven methods support this foundation, two distinctions keep them disciplined. The first separates a prediction task from a decision task: forecasting, state estimation, and option ranking can prepare a dispatch decision, but switching, congestion response, and reserve commitment remain safety- and liability-sensitive decisions that require explicit authority boundaries and guardrails. The second treats model operations as a living discipline rather than a one-off build. Models age, sensors drift, and data regimes shift, so retraining, drift detection, version control, and the lineage and reproducibility of the underlying data are operating obligations rather than project artefacts. Maturity in this layer is measured by whether data, models, decisions, safety logic, and security are held together in a governable operating architecture, not by the number of models or dashboards in use.

8.5 Operating model and organisational design

A flexibility-capable portfolio requires an organisation that connects market, technology, data, and settlement functionally rather than placing them side by side. A robust operating model typically encompasses several clearly distinguishable functional spaces.

Origination develops projects, market pathways, and contractual spaces. Asset management is responsible for technical performance, availability, and lifecycle management. Market operations organise schedules, bids, activations, and market communication. Trading and optimisation manage price exposure, allocation, and opportunities. Risk and compliance secure limits, counterparties, regulatory adherence, and escalation capability. Settlement and finance are responsible for settlement, liquidity, collateral, and result attribution. IT, data, and cyber ensure data availability, integration capability, and security. Commercial management connects these layers with strategic priorities and capital allocation.

Maturity does not arise from the formal presence of all functions, however. It reveals itself in the quality of their interfaces, as set out in Table 3. Where does technical dispatch end and market allocation begin? Who decides when degradation protection and short-term revenue conflict? Who holds the balancing group logic together? Who carries responsibility for data integrity, baseline methodology, and auditability? At precisely these interfaces, the load-bearing capacity of the operating model is determined.

Function	Minimum capability	Maturity indicator	Principal risk
Forecasting	Robust forecasting	Sustained forecast accuracy	Costly reactivity
Scheduling	Market- and asset-aligned scheduling	Consistent schedules	Misallocation
Dispatch	Real-time control capability	Low deviation from target	Activation errors
ETRM	Position and risk control	Transparency of exposures	Opacity
Validation	Traceable evidence base	High audit resilience	Dispute / non-remuneration
Governance	Clear rights and limits	Escalation-ready structure	Opportunism / conflict

Table 3: The operational foundations of a flexibility-capable portfolio – forecasting and dispatch through to governance and validation

8.6 KPI logic and operational transparency

As market and asset complexity grows, the risk increases that degrees of freedom are being used without being precisely understood. A robust operating model requires a KPI logic that translates technical, commercial, and operational performance into decision-relevant transparency.

Central metrics typically include technical availability, usable degrees of freedom, activation windows, SoC quality, reserve holdings, revenue structure by market and product logic, balancing deviations, forecast quality, activation and fulfilment rates, degradation indicators, counterparty exposure, liquidity and collateral load, and the opportunity costs of unused degrees of freedom.

The quality of a KPI system lies in whether it improves decisions rather than in the number of metrics it tracks. A portfolio that cannot make its degrees of freedom visible cannot govern them with discipline. Transparency is a precondition of operational controllability rather than a reporting question.

8.7 Development logic and maturity levels

Few organisations begin with a fully integrated flexibility model. Maturity develops incrementally in practice. A coherent development logic can be described across four levels.

- Level one is the market- and contract-capable asset, where grid connection, forecasting, schedule capability, and basic controllability are the primary focus.
- Level two is the optimisation-capable portfolio, where storage, flexible loads, intraday management, and balancing group logic are added.
- Level three is the flexibility-capable operating model, where balancing energy, local markets, re-optimisation, reserve management, governance, and limit logic gain in weight. Level four is the scalable platform and ETRM model, at which larger portfolios, multiple contract forms, multiple markets, more complex counterparty structures, and higher operational density can be robustly integrated.

These levels are not a rigid roadmap. They make visible, however, that flexibility arises through the densification of operational governability rather than through the addition of individual assets.

8.8 Operational conclusions

The operating and portfolio layer determines whether flexibility remains a theoretical action space or translates into a robust value architecture. Technical potentials and market connections are necessary but not sufficient. State awareness, prioritisation, risk management, ETRM, governance, measurement, validation, and settlement together constitute an operational practice that remains consistent under uncertainty.

The maturity of a flexibility model is measured by the discipline with which technical, market, contractual, and operational degrees of freedom are governed rather than by the number of markets addressed. This is precisely where the boundary runs between opportunistic market participation and a durable portfolio architecture.

9. Conclusions

The development of renewable portfolios is taking place in an environment where technical, market, and operational questions can no longer be meaningfully separated. As volatile generation expands, electrification deepens, network constraints become more structural, and flexibility and trading markets differentiate further, the logic by which assets are valued changes fundamentally. Whether a project can be built, connected, and placed on a bankable marketing pathway remains necessary. Increasingly decisive is whether its technical, contractual, and operational structure preserves those degrees of freedom that remain genuinely usable across changing market, network, and state conditions.

This is the deeper meaning of flexibility. It is the capacity to change behaviour deliberately, embedded within a robust action space – distinct from any single market product and irreducible to storage technology alone. That action space arises where physical degrees of freedom, market and contractual spaces, and operational governability converge into a coherent architecture.

The Asset and Engineering Layer has established that degrees of freedom take very different forms. PV and wind carry strong system impact whilst offering limited symmetric own flexibility. BESS opens fast, bidirectional, and high-precision degrees of freedom within the conditions of state management, ageing discipline, and careful dispatch. Electrolysis and hydrogen extend the freedom space towards controllable demand and sectoral conversion. Hydro, heat pumps, EV charging infrastructure, and data centres each add further, distinctively structured load, storage, and control logics. Flexibility is not a homogeneous resource but a technically differentiated, state-dependent capability that is always bound to its specific application.

The Market and Contract Layer has demonstrated that these degrees of freedom acquire economic relevance only when embedded in appropriate coordination spaces. Day-ahead and intraday markets, balancing energy markets, and local and vertical flexibility markets address different strata of the same system. Direct marketing, route-to-market, PPA, CPPA, and CfD structures redistribute price, volume, and profile risks whilst simultaneously preserving or curtailing degrees of freedom. Flexibility neither constitutes a detached speciality market nor a simple extension of wholesale trading. It is part of a graduated market and coordination order in which different mechanisms make different value dimensions legible.

The Operating and Portfolio Layer has made the actual maturity criterion visible. Degrees of freedom become economically valuable not by virtue of being theoretically addressable but through being operationally governable under uncertainty. Forecasting, scheduling, real-time management, hedging, ETRM, governance, measurement, validation, and settlement together form the architecture through which technical possibility is translated into durable operational capability. The decisive operational question is therefore not how many markets a portfolio could theoretically address, but with what discipline its degrees of freedom are prioritised, bounded, linked, and delimited under conflict conditions.

A clear consequence follows for developers, operators, IPPs, aggregators, and integrated platform and portfolio actors. Projects should be read as building blocks of a future action space – one in which grid connection, topology, storage sizing, load and conversion logic, contract structure, balancing group management, market access, platform capability, and data architecture together determine future behaviour. Thinking these relationships through early raises not only the marketability of an individual asset but the robustness of the entire portfolio.

For PV-and-BESS, wind-and-BESS, and hydrogen-and-BESS configurations this represents a strategic shift. The value of such systems arises from the quality of their coupling: between generation and storage, between load and network signal, between contract and optionality, between market and operating model, between data and decision. Technical complexity creates no additional value. Value arises where degrees of freedom are translated into an ordered, robust, and economically coherent management logic.

The purpose of this compendium can accordingly be stated precisely. It is not another survey of flexibility markets, storage technologies, or new energy services. It is an integrated systems perspective on how flexible renewable portfolios must be developed, contractually structured, marketed, and operationally governed so that technical changeability becomes durable economic and systemic value. In this sense, flexibility is one of the operational and commercial core conditions of the energy transition.

Annex I – Mathematical definitions and key inequalities

I.1 System boundaries, variables, and state descriptions

I.1.1 Fundamental system form

The flexibility system considered in this compendium is described as a time-discrete, state-dependent decision and control system. Its behaviour arises from the coupling of physical assets, market and contractual commitments, exogenous influences, and operational control decisions.

Over a finite time horizon $\mathcal{T} = \{0, 1, \dots, T\}$, let for each $t \in \mathcal{T}$:

- $x_t \in \mathcal{X}$ denote the physical state vector,
- $z_t \in \mathcal{Z}$ the market and contractual state vector,
- $u_t \in \mathcal{U}$ the vector of controllable decisions,
- $w_t \in \mathcal{W}$ the vector of exogenous influences.

The overall system is described by the augmented state vector $s_t := (x_t, z_t)$. System dynamics are given by

$$x_{t+1} = f(x_t, u_t, w_t), \quad z_{t+1} = q(z_t, u_t, m_t), \quad \text{where } m_t \text{ denotes market- or contract-specific events, activations, or clearing outcomes.}$$

In compact form, $s_{t+1} = F(s_t, u_t, \omega_t)$, where ω_t collects all exogenous influences. This formulation is sufficiently general to describe generation, storage, flexible loads, market positions, reservations, balancing, and activation within a single framework.

I.1.2 Time discretisation and scale levels

Time is partitioned into intervals of length Δt , with each interval t mapping to $t \mapsto [t\Delta t, (t+1)\Delta t)$.

The choice of Δt is constitutive for the model. It determines which degrees of freedom become visible and which dynamics appear smoothed or exaggerated. At least three-time levels are relevant for this compendium:

1. Planning level: day-ahead commitments, contract allocation, capacity reservation.
2. Adjustment level: intraday re-optimisation, state updating.
3. Activation level: real-time dispatch, balancing energy, local activation.

These levels may be discretised at different granularities. The embedding of finer into coarser time scales can be described formally through mappings between the respective time sets. The essential point is that a degree of freedom appearing available on a coarse planning scale may already be committed, through state conditions or reservations, at a finer activation scale.

I.1.3 Power, energy, and sign convention

For asset $i \in \mathcal{I}$, let $P_{i,t}$ denote the active power in interval t . The corresponding energy quantity is $E_{i,t} = P_{i,t}\Delta t$.

The sign convention is set consistently as follows:

- $P_{i,t} > 0$: injection into the electrical system under consideration,
- $P_{i,t} < 0$: withdrawal from the electrical system under consideration.

For bidirectional assets, in particular storage, decomposition into charge and discharge power is appropriate:

$$P_{i,t} = P_{i,t}^{dis} - P_{i,t}^{ch}, \quad P_{i,t}^{dis} \geq 0, \quad P_{i,t}^{ch} \geq 0.$$

The net portfolio power is $P_t^{port} = \sum_{i \in \mathcal{I}} P_{i,t}$.

Power and energy must be kept strictly distinct. High power flexibility implies neither long duration nor large energy content; large energy content implies no high short-term response capability.

I.1.4 State space

A degree of freedom is always state-dependent. For each asset i , a state vector $x_{i,t} \in \mathcal{X}_i$ is introduced. The aggregated physical state space is $x_t = (x_{i,t})_{i \in \mathcal{I}} \in \mathcal{X} := \prod_{i \in \mathcal{I}} \mathcal{X}_i$.

Typical state variables include:

- state of charge of a storage system,
- thermal state of a building or heat storage,
- operating mode of an electrolyser,
- availability state of a plant,
- reserved power or state windows,
- local network restrictions.

State evolution follows generally $x_{t+1} = f(x_t, u_t, w_t)$.

I.1.5 Market and contractual state

In addition to the physical state, a market and contractual state space $z_t \in \mathcal{Z}$ is introduced. Typical components include:

- day-ahead positions,
- intraday corrections,
- balancing energy reservations,
- local activation commitments,

- PPA, CPPA, or CfD-related obligations,
- balancing group positions,
- available or reserved degrees of freedom.

The dynamics of this state space are governed $z_{t+1} = q(z_t, u_t, m_t)$.

A degree of freedom may therefore still be physically present whilst already being committed in market or contractual terms. This is precisely why the combined state $s_t = (x_t, z_t)$ is central to the subsequent modelling.

I.1.6 Control variables and degrees of freedom

Controllable decisions are described by $u_t \in \mathcal{U}(s_t, \theta_t)$, where θ_t collects technical, contractual, market, and regulatory parameters. Typical components of u_t include:

- charge and discharge power,
- curtailment,
- load increase or load reduction,
- activation bids,
- reservation decisions,
- schedule adjustments,
- internal reallocation of degrees of freedom.

The degree of freedom at time t is thus formally the admissible decision set $\mathcal{F}_t := \mathcal{U}(s_t, \theta_t)$.

It is not a fixed characteristic but a time-, state-, and parameter-dependent action space.

I.1.7 Constraints

The system's action space is bounded by equality and inequality constraints:

$$g(s_t, u_t, w_t) \leq 0, \quad h(s_t, u_t, w_t) = 0.$$

The inequalities g describe, for example:

- power limits,
- energy and SoC bounds,
- ramp rates,
- minimum loads,
- comfort and process bounds,
- grid connection limits,
- reservations,
- contractual obligations,
- non-simultaneity conditions.

The equalities h describe in particular:

- state transitions,
- energy balances,
- schedule and delivery equations,
- balancing group relationships.

Unmodelled constraints artificially enlarge the action space. The formal quality of the model therefore depends substantially on the completeness and relevance of this constraint set.

I.1.8 Locational reference

For local and vertical flexibility markets, a location index $n \in \mathcal{N}$ is introduced, describing a grid connection point, node, network section, or zone depending on the modelling purpose.

Power and states may then be written location-dependently as: $P_{n,t}, x_{n,t}$.

This makes visible that degrees of freedom are structured spatially as well as temporally. For local markets, it is precisely this spatial structure that is value-determining.

I.1.9 Exogenous processes

The exogenous influences w_t encompass in particular:

- weather,
- prices,
- network signals,
- load behaviour,
- activation calls,
- regulatory parameters,
- comfort or production requirements.

They may be modelled deterministically, scenario-based, or stochastically. In the stochastic case, $w_t \sim P_t$ is a random variable with time-dependent distribution P_t .

I.1.10 Objective function

The general objective function of a flexibility-capable portfolio is multi-dimensional:

$$\max_{u_t} \sum_{t \in T} (R_t^{energy} + R_t^{balancing} + R_t^{local} + R_t^{contract} - C_t^{imb} - C_t^{deg} - C_t^{cong} - C_t^{risk}).$$

This establishes the formal foundation: technical degrees of freedom are understood as dynamic, state- and parameter-dependent decision spaces under constraints, whose economic value arises only through the interplay of markets, contracts, and operational management.

I.2 Asset-specific constraint sets

I.2.1 Starting point

The general form of Section I.1 describes the structure of the system rather than the asset-specific form of its degrees of freedom. For each asset $i \in \mathcal{I}$, a specific set of states, control variables, and constraints is therefore introduced:

$$g_i(x_{i,t}, u_{i,t}, w_{i,t}) \leq 0, \quad h_i(x_{i,t}, u_{i,t}, w_{i,t}) = 0.$$

These constraints define the actual action space of the asset and thereby its real contribution to market participation, reservation, dispatch, and portfolio optimisation.

I.2.2 Photovoltaics

For PV, let P_t^{PV} denote the injected active power. It holds that $0 \leq P_t^{PV} \leq \bar{P}_t^{irr}$, where $\bar{P}_t^{irr}$ denotes the maximum available power resulting from irradiance, temperature, module configuration, shading, and inverter settings.

With curtailment control variable $u_t^{curt,PV} \geq 0$: $P_t^{PV} = \bar{P}_t^{irr} - u_t^{curt,PV}$.

Additional constraints: $P_t^{PV} \leq \bar{P}^{grid,PV}$, $|P_t^{PV} - P_{t-1}^{PV}| \leq R^{PV}$.

With explicit reactive power modelling: $(P_t^{PV})^2 + (Q_t^{PV})^2 \leq (\bar{S}^{PV})^2$.

PV is thus an asymmetrically controllable, resource-dependent asset: reduction is feasible; additional positive power without storage is not.

I.2.3 Wind

For wind: $0 \leq P_t^{Wind} \leq \bar{P}_t^{wind}$, with $\bar{P}_t^{wind}$ as the resource-dependent upper bound.

With curtailment or de-loading variable: $P_t^{Wind} = \bar{P}_t^{wind} - u_t^{curt,Wind}$, $u_t^{curt,Wind} \geq 0$.

Additional constraints: $|P_t^{Wind} - P_{t-1}^{Wind}| \leq R^{Wind}$, $P_t^{Wind} \leq \bar{P}^{grid,Wind}$, $(P_t^{Wind})^2 + (Q_t^{Wind})^2 \leq (\bar{S}^{Wind})^2$.

For wind farms: $P_t^{Wind} = \sum_{j \in \mathcal{J}} P_{j,t}^{Wind}$.

Wind carries primarily negative flexibility, as does PV, yet typically with stronger relevance on the transmission and load-flow side.

I.2.4 Battery energy storage systems

For a BESS, charge and discharge power are modelled separately: $P_t^{ch} \geq 0$, $P_t^{dis} \geq 0$.

Net power: $P_t^{BESS} = P_t^{dis} - P_t^{ch}$.

Power limits: $0 \leq P_t^{ch} \leq \bar{P}^{ch}$, $0 \leq P_t^{dis} \leq \bar{P}^{dis}$.

Non-simultaneity, MILP-compatible with $y_t^{BESS} \in \{0,1\}$: $P_t^{ch} \leq \bar{P}^{ch}(1 - y_t^{BESS})$, $P_t^{dis} \leq \bar{P}^{dis} y_t^{BESS}$.

SoC dynamics: $SoC_{t+1} = SoC_t + \eta^{ch} P_t^{ch} \Delta t - \frac{1}{\eta^{dis}} P_t^{dis} \Delta t$.

Bounds: $\underline{SoC} \leq SoC_t \leq \overline{SoC}$.

With state of health SoH_t : $SoH_{t+1} = SoH_t - d_t$,

where d_t is a degradation function: $d_t = \psi(P_t^{ch}, P_t^{dis}, SoC_t, T_t, DoD_t)$.

Additional constraints: $\frac{P_t^{ch}}{E^{nom}} \leq C^{ch}$, $\frac{P_t^{dis}}{E^{nom}} \leq C^{dis}$, $T_t^{min} \leq T_t \leq T_t^{max}$.

A BESS is thus a bidirectional, state- and ageing-sensitive storage system with explicit energy-state dynamics.

I.2.5 Elektrolyser and hydrogen pathway

For the electrolyser, let P_t^{el} denote the electrical power intake with binary operating variable $y_t^{el} \in \{0,1\}$:

$$\underline{P}^{el} y_t^{el} \leq P_t^{el} \leq \bar{P}^{el} y_t^{el}.$$

Hydrogen production: $H_t = \eta_t^{el} P_t^{el} \Delta t$, or more generally with load-dependent efficiency: $H_t = \eta_t^{el}(P_t^{el}) P_t^{el} \Delta t$.

With hydrogen storage S_t^{H2} : $S_{t+1}^{H2} = S_t^{H2} + H_t - O_t^{H2}$, $\underline{S}^{H2} \leq S_t^{H2} \leq \bar{S}^{H2}$.

Additional constraints: $|P_t^{el} - P_{t-1}^{el}| \leq R^{el}$, together with start/stop logic and peripheral, compression, and offtake constraints.

The electrolyser is thus a controllable load with minimum load, ramp constraints, and process coupling.

I.2.6 Heat pump

For the heat pump, let P_t^{HP} denote the electrical power intake and Q_t^{HP} the thermal output:

$$Q_t^{HP} = COP_t \cdot P_t^{HP}.$$

With thermal state T_t : $T_{t+1} = aT_t + bQ_t^{HP} - dL_t$, where L_t describes the thermal load.

Comfort or process bounds: $\underline{T} \leq T_t \leq \bar{T}$.

Power limit: $0 \leq P_t^{HP} \leq \bar{P}^{HP}$.

The degree of freedom here lies in the thermal state trajectory rather than in unconstrained electrical variability.

I.2.7 EV charging infrastructure

For EV charging infrastructure: $0 \leq P_t^{EV} \leq \bar{P}^{EV}$.

For a charging window $[t_a, t_d]$ with required energy E^{req} : $\sum_{t=t_a}^{t_d} P_t^{EV} \Delta t \geq E^{req}$.

Optionally with vehicle-internal SoC: $SoC_{t+1}^{EV} = SoC_t^{EV} + \eta^{EV} P_t^{EV} \Delta t$, $\underline{SoC}^{EV} \leq SoC_t^{EV} \leq \overline{SoC}^{EV}$.

The degree of freedom lies in the time window and aggregability rather than in unlimited power freedom.

I.2.8 Data centre

The total load of a data centre is: $P_t^{DC} = P_t^{IT} + P_t^{cool} + P_t^{aux}$. The IT core load is typically only limitedly dispatchable.

Degrees of freedom lie principally in cooling, auxiliary systems, backup architecture, and partial workload shifting.

With server temperature T_t^{srv} : $T_{t+1}^{srv} = aT_t^{srv} + bP_t^{IT} - cQ_t^{cool}$, $\underline{T}^{srv} \leq T_t^{srv} \leq \bar{T}^{srv}$.

With cooling limits: $0 \leq Q_t^{cool} \leq \bar{Q}^{cool}$.

Data centres are high-energy, digitally well-manageable loads with narrow but precisely usable degree-of-freedom corridors.

I.2.9 Hydro

For hydro, let S_t^{hyd} denote the reservoir level: $S_{t+1}^{hyd} = S_t^{hyd} + I_t + P_t^{pump} \Delta t - O_t - P_t^{tur} \Delta t$, with inflow I_t , spillage O_t , pumping power P_t^{pump} and turbine output P_t^{tur} .

Bounds: $\underline{S}^{hyd} \leq S_t^{hyd} \leq \bar{S}^{hyd}$, $0 \leq P_t^{tur} \leq \bar{P}^{tur}$, $0 \leq P_t^{pump} \leq \bar{P}^{pump}$.

Hydro couples' power and duration differently from BESS and is typically more strongly shaped by site conditions, inflow regimes, and environmental constraints.

I.2.10 Aggregated loads and demand-side flexibility

For an aggregated load portfolio $\mathcal{L}$: $P_t^{load,agg} = \sum_{l \in \mathcal{L}} P_{l,t}$.

With baseline P_t^{base} , the usable demand-side flexibility is:

$$\Delta P_t^{DSF} = P_t^{base} - P_t^{load,agg}, \text{ with upper limit } \Delta P_t^{DSF} \leq \sum_{l \in \mathcal{L}} \bar{F}_{l,t}.$$

The challenge here lies less in the individual plant than in heterogeneity, synchronisation, baseline logic, and exclusivity.

I.2.11 Portfolio-wide coupling constraints

At portfolio level, competition and coupling between degrees of freedom become visible.

Grid connection limit: $\sum_{i \in \mathcal{J}^{exp}} P_{i,t} - \sum_{j \in \mathcal{J}^{imp}} P_{j,t} \leq \bar{P}^{grid}$.

Non-double-marketing: $u_t^{energy} + u_t^{bal} + u_t^{local} \leq \bar{u}_t^{flex}$.

Reserve holding: $SoC_t \geq SoC_t^{reserve}$, or more generally $x_t \in \mathcal{X}^{reserve}$.

Contractual obligation: $q_t^{delivery} \geq q_t^{contract}$, or under strict profile binding $q_t^{delivery} = q_t^{contract}$.

These constraints are frequently more decisive than individual asset constraints. They reveal whether a portfolio is genuinely flexible or merely apparently versatile.

I.2.12 Structural conclusion

Asset-specific constraints demonstrate that flexibility is not a homogeneous resource but a family of structurally distinct decision spaces. The differences lie less in the semantic description of asset types than in:

- the structure of their admissible sets $\mathcal{F}_{i,t}$,
- the form of their state dynamics,
- the nature of their coupling constraints, and
- the way they can be embedded in portfolio and market logics.

Precisely from these differences arise the distinctions in reservation capability, activatability, hedging function, marketability, and governance requirements.

1.3 Market and dispatch logic

1.3.1 Starting point

Technical degrees of freedom become economically effective only when translated into market and activation logics. These logics differ in time horizon, product definition, reservation requirements, locational specificity, remuneration form, and settlement methodology. A mathematical framework must therefore formalise not only technical states but also market application spaces.

1.3.2 Day-ahead decision

For each time slot t a day-ahead position q_t^{DA} is established. The associated revenue is $R^{DA} = \sum_{t \in \mathcal{T}} \pi_t^{DA} q_t^{DA} \Delta t$.

The admissible position depends not only on forecast generation or load but equally on storage states, reservations, grid connection limits, contractual obligations, and risk limits.

1.3.3 Intraday adjustment

With intraday correction q_t^{ID} , the final marketed position is $q_t^{MKT} = q_t^{DA} + q_t^{ID}$.

Intraday revenue: $R^{ID} = \sum_{t \in \mathcal{T}} \pi_t^{ID} q_t^{ID} \Delta t$.

The economic function of the intraday market lies in forecast correction, balancing group smoothing, and opportunistic re-optimisation alongside supplementary trading.

1.3.4 Balancing deviation and imbalance logic

With actual behaviour P_t^{real} , the balancing deviation is: $\epsilon_t = P_t^{real} - q_t^{MKT}$.

Imbalance costs: $C^{imb} = \sum_{t \in \mathcal{T}} (\lambda_t^+ \max(\epsilon_t, 0) + \lambda_t^- \max(-\epsilon_t, 0))$.

Balancing deviation is therefore not a residual variable but a central coupling point between forecasting, market position, and real-time behaviour.

1.3.5 Balancing energy: reservation and activation

For balancing energy, reservation and activation must be modelled separately.

With reserved capacity r_t^{cap} and activated power r_t^{act} :

$$0 \leq r_t^{act} \leq r_t^{cap}.$$

With separate upward and downward products: $r_t^{cap,+}$, $r_t^{cap,-}$, $r_t^{act,+}$, $r_t^{act,-}$, revenues are:

$$R^{bal} = \sum_{t \in \mathcal{T}} (\pi_t^{cap} r_t^{cap} + \pi_t^{act} r_t^{act}) \Delta t.$$

For a BESS, reservation directly implies SoC constraints: $SoC_t \geq \underline{SoC} + \alpha r_t^{cap,+}$, $SoC_t \leq \overline{SoC} - \beta r_t^{cap,-}$.

Balancing energy is thus not a frictionless additional market but alters the admissible state space of the portfolio directly.

1.3.6 Local flexibility markets

With location index n and local activation $l_{n,t}$, the local value term is: $R^{local} = \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{T}} \pi_{n,t}^{local} l_{n,t} \Delta t$.

Activation is subject to location-specific constraints: $0 \leq l_{n,t} \leq \bar{l}_{n,t}$, $g_{n,t}^{grid}(P_{n,t}, l_{n,t}) \leq 0$.

Local markets differ formally from wholesale markets principally through the explicit location index and a value logic oriented towards network relief or location-specific effect.

1.3.7 Vertical coordination

Vertical coordination means formally the simultaneous consideration of multiple value levels:

$$\max \sum_t (R_t^{energy} + R_t^{balancing} + R_t^{local}) - C_t^{risk}$$

$$\text{subject to the coupling constraint } u_t^{energy} + u_t^{balancing} + u_t^{local} \leq \bar{u}_t.$$

A degree of freedom can therefore be committed only once. Its economic value arises from the choice between competing applications.

1.3.8 Contractual obligation and dispatch

With contractual target profile $q_t^{contract}$ and actual delivery $q_t^{delivery}$ a deviation penalty can be formulated as:

$$C^{contract} = \sum_t \lambda_t^{contract} |q_t^{delivery} - q_t^{contract}|.$$

Alternatively, a hard constraint may apply: $q_t^{delivery} = q_t^{contract}$.

This choice is not a mathematical detail but reflects the actual stringency of the contractual logic.

1.3.9 Dispatch as the realisation level

With planned decision u_t^{plan} and actual implementation u_t^{real} , the operative deviation is:

$$\delta_t = u_t^{real} - u_t^{plan}.$$

Dispatch is thus the final realisation stage in a chain of forecasting, market decision, and activation. Models must be robust enough to deliver economically sound decisions even when $\delta_t \neq 0$.

1.3.10 Opportunity costs

Where a degree of freedom is deployed in market or function A it is only partially available for B or C .

The opportunity value is: $OC_t = \max\{V_t^A, V_t^B, V_t^C, \dots\} - V_t^{chosen}$.

This formulation is central for flexibility portfolios. Every activation carries not only a revenue but also a foregone value.

I.3.11 Multi-market general form

A general multi-market formulation:

$$\max \sum_{t \in \mathcal{T}} (R_t^{DA} + R_t^{ID} + R_t^{bal} + R_t^{local} - C_t^{imb} - C_t^{deg} - C_t^{contract} - C_t^{risk})$$

subject to $x_{t+1} = f(x_t, u_t, w_t)$, $u_t \in \mathcal{U}(x_t, \theta_t)$, and all asset- and portfolio-wide constraints.

The central challenge lies not in the existence of individual markets but in the coupled management of the same degrees of freedom across multiple markets simultaneously.

I.4 Portfolio and Hedging

I.4.1 Starting point

With the transition from asset to portfolio, the mathematical perspective shifts. Degrees of freedom are no longer evaluated in isolation but relative to revenue spaces, obligations, risks, and alternative uses. Hedging in this context is not mere price protection but the structuring function through which uncertainty is translated into a governable portfolio space.

I.4.2 Portfolio value

The portfolio value at time t is $V_t = V_t^{energy} + V_t^{balancing} + V_t^{local} + V_t^{contract} + V_t^{optional} - C_t$.

Over the horizon: $V = \sum_{t \in \mathcal{T}} V_t$.

The components are not independent. A degree of freedom deployed in market A today may reduce availability in market B tomorrow. Portfolio value is therefore a coupled function of competing applications.

I.4.3 Price risk

For market m let $\pi_{m,t}$ denote the price process and $q_{m,t}$ the associated position. Then:

$$R_m = \sum_{t \in \mathcal{T}} \pi_{m,t} q_{m,t} \Delta t.$$

Price risk arises from the uncertainty of R_m , expressed for instance through $\text{Var}(R_m)$ or, across the portfolio $\text{Var}(\sum_m R_m)$. Spreads, volatilities, and correlations are as relevant as absolute price levels.

I.4.4 Volume risk

With forecast quantity $\hat{P}_t$ and volume error ϵ_t^{vol} :

$$P_t^{real} = \hat{P}_t + \epsilon_t^{vol}, \quad \Delta Q_t = Q_t^{real} - Q_t^{plan}.$$

Volume costs: $C_t^{vol} = \lambda_t^+ \max(\Delta Q_t, 0) + \lambda_t^- \max(-\Delta Q_t, 0)$.

Volume risk is particularly relevant for weather-dependent assets, state-constrained storage systems, and flexible loads with usage-dependent availability.

I.4.5 Profile risk

With asset or portfolio profile $q_t^{profile}$ and target profile d_t^{target} the profile deviation is:

$$\Delta_t^{shape} = q_t^{profile} - d_t^{target}.$$

A simple cost function: $C^{shape} = \sum_{t \in \mathcal{T}} \lambda_t^{shape} |\Delta_t^{shape}|$.

Shape risk is of high relevance for PPA and CPPA structures, profile smoothing obligations, and hybrid portfolios.

I.4.6 Imbalance risk

With planned position q_t^{plan} and actual behaviour q_t^{real} : $\epsilon_t = q_t^{real} - q_t^{plan}$.

Imbalance costs: $C^{imb} = \sum_{t \in \mathcal{T}} (\lambda_t^{imb,+} \max(\epsilon_t, 0) + \lambda_t^{imb,-} \max(-\epsilon_t, 0))$.

Imbalance risk is closely coupled to forecasting quality, intraday management, and SoC or load management discipline.

I.4.7 Congestion risk

With network-side constraint $g_{n,t}^{grid}(P_{n,t}) \leq 0$, the cost effect of congestion can be written in simplified form as:

$$C^{cong} = \sum_{n,t} \lambda_{n,t}^{cong} \max(g_{n,t}^{grid}(P_{n,t}), 0).$$

Alternatively, local activation can create additional value. Alter: $R^{local} = \sum_{n,t} \pi_{n,t}^{local} l_{n,t} \Delta t$.

Congestion risk thus carries a dual character: it is simultaneously a risk and a source of opportunity.

I.4.8 Degradation risk

For a BESS or other ageing-sensitive asset, a simple linear degradation cost function is:

$$C^{deg} = \sum_{t \in \mathcal{T}} c^{deg} (P_t^{ch} + P_t^{dis}) \Delta t.$$

More precisely: $C_t^{deg} = f(DoD_t, C_t, T_t, SoC_t)$.

Degradation belongs inside the objective function rather than at its margin. A portfolio can achieve nominally high revenues whilst destroying long-term value if ageing costs remain implicit.

1.4.9 Hedge ratio

With open exposure E_t and hedge position H_t : $h_t = \frac{H_t}{E_t}$.

In the flexibility context, the hedge may be physical, contractual, or financial. A storage asset, an electrolyser, or a PPA can each fulfil a hedge function.

1.4.10 Correlation and portfolio effect

For two value contributions R_1 and R_2 : $\text{Var}(R_1 + R_2) = \text{Var}(R_1) + \text{Var}(R_2) + 2\text{Cov}(R_1, R_2)$.

Generalised to N components: $\text{Var}(\sum_{i=1}^N R_i) = \sum_{i=1}^N \text{rVar}(R_i) + 2 \sum_{i < j} \text{rCov}(R_i, R_j)$.

The added value of hybrid portfolios arises substantially from the modification of this covariance structure.

1.4.11 VaR and CVaR

With loss variable L , the Value-at-Risk at confidence level α is defined by: $P(L \leq \text{VaR}_\alpha) =$.

The Conditional Value-at-Risk: $\text{CVaR}_\alpha = E[L \mid L \geq \text{VaR}_\alpha]$.

For flexibility portfolios, the CVaR is frequently more informative than the VaR, since extreme states are decisive for actual robustness.

1.4.12 Risk-adjusted objective function

A general risk-adjusted objective function is:

$$\max_{\sum_{t \in T}} (R_t^{\text{energy}} + R_t^{\text{bal}} + R_t^{\text{local}} + R_t^{\text{contract}} - C_t^{\text{imb}} - C_t^{\text{deg}} - C_t^{\text{cong}}) - \lambda_1 \text{Var}(V) - \lambda_2 \text{CVaR}.$$

Hedging thus appears as an architectural discipline rather than merely a trading function.

1.5 Multi-market co-optimisation

1.5.1 Starting point

Once a portfolio simultaneously addresses wholesale markets, balancing energy, local flexibility markets, contractual obligations, and internal operating objectives, sequential single-market optimisation is insufficient. The same degrees of freedom are claimed across multiple application spaces. The mathematical response is co-optimisation.

1.5.2 General form

With control vector u_t , the problem is:

$$\begin{aligned} \max_{u_t} \sum_{t \in T} & (R_t^{\text{DA}} + R_t^{\text{ID}} + R_t^{\text{bal}} + R_t^{\text{local}} + R_t^{\text{contract}} - C_t^{\text{imb}} - C_t^{\text{deg}} - C_t^{\text{cong}} - C_t^{\text{risk}}) \\ \text{subject to } & x_{t+1} = f(x_t, u_t, w_t), \quad u_t \in \mathcal{U}(x_t, \theta_t), \quad g(x_t, u_t, w_t) \leq 0, \quad h(x_t, u_t, w_t) = 0. \end{aligned}$$

1.5.3 Market segments

The value contributions can be written as:

$$R_t^{\text{DA}} = \pi_t^{\text{DA}} q_t^{\text{DA}} \Delta t, \quad R_t^{\text{ID}} = \pi_t^{\text{ID}} q_t^{\text{ID}} \Delta t, \quad R_t^{\text{bal}} = \pi_t^{\text{cap}} r_t^{\text{cap}} \Delta t + \pi_t^{\text{act}} r_t^{\text{act}} \Delta t, \quad R_t^{\text{local}} = \pi_{n,t}^{\text{local}} l_{n,t} \Delta t.$$

The difficulty lies not in the existence of individual terms but in their coupling.

1.5.4 Allocation constraint

A degree of freedom can be committed only once. In general: $u_t^{\text{DA}} + u_t^{\text{ID}} + u_t^{\text{bal}} + u_t^{\text{local}} + u_t^{\text{contract}} \leq \bar{u}_t$.

For a BESS, this can be formulated separately for upward and downward directions:

$$P_t^{\text{dis}} \geq q_t^{\text{DA},+} + q_t^{\text{ID},+} + r_t^{\text{cap},+} + l_t^+, \quad P_t^{\text{ch}} \geq q_t^{\text{DA},-} + q_t^{\text{ID},-} + r_t^{\text{cap},-} + l_t^-.$$

1.5.5 State-dependent co-optimisation

Co-optimisation is state bound. For a BESS with reservation:

$$\text{SoC}_{t+1} = \text{SoC}_t + \eta^{\text{ch}} P_t^{\text{ch}} \Delta t - \frac{1}{\eta^{\text{dis}}} P_t^{\text{dis}} \Delta t, \quad \text{SoC}_t \geq \underline{\text{SoC}} + \alpha r_t^{\text{cap},+}, \quad \text{SoC}_t \leq \overline{\text{SoC}} - \beta r_t^{\text{cap},-}.$$

Reservation thus alters the admissible state space directly.

1.5.6 Hierarchical co-optimisation

Where markets or functions are not of equal priority, a weighted or lexicographic structure can be adopted.

Weighted form:

$$\max_{\sum_t} (W_1 R_t^{\text{critical}} + W_2 R_t^{\text{bal}} + W_3 R_t^{\text{energy}} + W_4 R_t^{\text{local}}) - C_t \quad \text{with } W_1 \gg W_2 \gg W_3 \gg W_4.$$

Lexicographic form: $\max R^{(1)} \text{ s.t. } \mathcal{C}, \max R^{(2)} \text{ s.t. } \mathcal{C}, R^{(1)} = R^{(1)*}, \text{ etc.}$

This structure is frequently closer to operational reality than simultaneous maximisation across all revenue spaces.

1.5.7 Contractual obligation

With contractual target profile q_t^{target} and delivered quantity q_t^{delivery} :

$$C_t^{\text{contract}} = \lambda_t^{\text{contract}} |q_t^{\text{delivery}} - q_t^{\text{target}}|, \quad \text{or as a hard constraint: } q_t^{\text{delivery}} = q_t^{\text{target}}.$$

The stringency of the contractual world is thereby embedded directly in the optimisation space.

1.5.8 Local markets

With location index n : $R^{local} = \sum_{n,t} \pi_{n,t}^{local} l_{n,t} \Delta t$, subject to $g_{n,t}^{grid}(P_{n,t}, l_{n,t}) \leq 0$.

The objective conflict between wholesale trading, balancing energy, local market, and contractual obligation is thereby made explicit.

1.5.9 Exclusivity

Exclusive uses can be modelled with binary variables:

$$y_t^{bal} + y_t^{arb} + y_t^{local} \leq 1, \text{ with } y_t^{bal}, y_t^{arb}, y_t^{local} \in \{0,1\}. \text{ Or continuously: } r_t^{cap} + q_t^{arb} + l_t \leq \bar{P}_t^{flex}.$$

1.5.10 Rolling horizon

In practice, co-optimisation is typically solved on a rolling basis. For horizon H : $u_t^* = \arg \max_{u_{t:t+H}} V_{t:t+H}$.

After implementing the first decision, re-optimisation proceeds with updated states and forecasts. This is the mathematical counterpart of the forecasting, scheduling, and real-time management axis described in the main text.

1.5.11 Shadow prices

For a degree-of-freedom constraint such as $u_t^{DA} + u_t^{ID} + u_t^{bal} + u_t^{local} \leq \bar{u}_t$, the associated Lagrange multiplier measures the marginal value of additional flexibility. These shadow prices are often more informative for asset sizing, contract valuation, and portfolio architecture than nominal revenues alone.

1.5.12 Robust and stochastic co-optimisation

Stochastic form: $\max \sum_{s \in \mathcal{S}} p_s V^{(s)}$.

Robust form: $\max_u \min_{w \in \mathcal{W}} V(u, w)$.

Risk-adjusted form: $\max E[V] - \lambda_1 Var(V) - \lambda_2 CVaR_\alpha$.

Co-optimisation thus becomes the search for a combination of degrees of freedom that remains economically and operationally robust under uncertainty.

1.6 Baseline, validation, and settlement logic

1.6.1 Starting point

Once flexibility is activated or remunerated in market terms, technical change alone is no longer sufficient. What becomes decisive is whether that change can be referenced, measured, validated, and settled. Baseline, validation, and settlement form the bridge between physical behaviour and institutionally recognised performance.

1.6.2 Baseline

With baseline B_t and observed actual behaviour P_t^{real} , the activated flexibility is, in the simplest case for a load reduction $F_t = B_t - P_t^{real}$ or more generally, sign-consistent: $F_t = \sigma(P_t^{real} - B_t)$, $\sigma \in \{-1, +1\}$.

The baseline is thus the counterfactual reference rather than the measured behaviour itself.

1.6.3 Baseline types

Historical baseline: $B_t = \frac{1}{K} \sum_{k=1}^K P_t^{hist,k}$.

Model-based baseline: $B_t = f(z_t)$.

Schedule-based baseline: $B_t = q_t^{plan}$.

Contractually defined baseline: $B_t = q_t^{contract}$.

Every baseline carries an error term: $B_t = P_t^{counterfactual} + \varepsilon_t^B$.

A baseline is therefore never identical with truth but always an institutionally accepted estimate of counterfactual behaviour.

1.6.4 Measurement and validation

With measured quantity $\tilde{P}_t = P_t^{real} + \varepsilon_t^M$, the measured flexibility is: $\tilde{F}_t = B_t - \tilde{P}_t$.

Let validation be a checking operator $\mathcal{V}(\tilde{F}_t, B_t, \Gamma_t) = v_t \in [0,1]$. The validated flexibility is then: $F_t^{valid} = v_t \tilde{F}_t$.

The rule set Γ_t typically encompasses:

- activation windows,
- minimum power thresholds,
- data completeness requirements,
- consistency rules,
- exclusion of contradictory activations.

1.6.5 Transfer of energy

With validated energy shift $\Delta E_t = F_t^{valid} \Delta t$, a transfer function $\mathcal{T}(\Delta E_t) = (\Delta E_t^{(1)}, \dots, \Delta E_t^{(N)})$ assigns this shift to multiple roles or balancing spaces such that $\sum_{k=1}^N \Delta E_t^{(k)} = \Delta E_t$.

Transfer of energy determines to whom the physical change is attributed economically and in balancing terms.

I.6.6 Settlement

General payment: $S_t = \pi_t^{flex} F_t^{valid} \Delta t$.

With capacity and activation components: $S_t = \pi_t^{cap} \gamma_t^{cap} \Delta t + \pi_t^{act} F_t^{valid} \Delta t$.

With transfer-of-energy compensation: $S_t^{net} = \pi_t^{flex} F_t^{valid} \Delta t - \lambda_t^{ToE} \Delta E_t$.

Multi-party distribution: $S_t^{gross} = \sum_{k=1}^N S_t^{(k)}$, $S_t^{(k)} = \alpha_k S_t^{gross} - C_t^{(k)}$, $\sum_{k=1}^N \alpha_k = 1$.

I.6.7 Conflict cases

Double attribution: $F_t^{(1)} + F_t^{(2)} > \bar{F}_t$ is inadmissible and requires $F_t^{(1)} + F_t^{(2)} \leq \bar{F}_t$.

Baseline manipulation: $B_t^{manip} = B_t + \delta_t^{strategic}$.

Time synchronisation error: $B_t - \widetilde{P}_{t+\tau}$, $\tau \neq 0$.

Such inconsistencies frequently determine whether a market remains robust or becomes institutionally fragile.

I.6.8 Location-specific baselines

For local markets: $B_{n,t}$, $F_{n,t}^{valid} = v_{n,t}(B_{n,t} - \widetilde{P}_{n,t})$, $S_{n,t}^{local} = \pi_{n,t}^{local} F_{n,t}^{valid} \Delta t$.

Local markets thus require not only temporal and state reference but also locational reference.

I.6.9 Portfolio consistency

With aggregated baseline $B_t^{port} = \sum_{i \in J} B_{i,t}$ and validated portfolio flexibility $F_t^{port} = \sum_{i \in J} F_{i,t}^{valid}$, it must hold for each asset that: $\sum_{m \in M} F_{i,t}^{(m)} \leq \bar{F}_{i,t}$.

This prevents the same flexibility from appearing twice across multiple markets or settlement systems.

I.7 Incentives, liquidity, and market design conditions

I.7.1 Starting point

Technical flexibility becomes market-effective only where actors have a genuine incentive to provide it. Between physical potential and actual market participation lie entry costs, frictions, risk, minimum size thresholds, product compatibility, and regulatory conditions. This layer is not ancillary to the mathematics but determines whether a flexibility market is genuinely sustained.

I.7.2 Participation decision

For participant i the expected net benefit is: $U_i = E[R_i^{flex}] - C_i^{entry} - C_i^{oper} - C_i^{risk}$.

Participation occurs when $U_i \geq 0$.

With risk aversion ρ_i : $U_i = E[R_i^{flex}] - C_i^{entry} - C_i^{oper} - \rho_i \cdot \text{Risk}_i$.

Participation is thus the outcome of a complete benefit–cost–risk assessment.

I.7.3 Minimum size and aggregation

With technical degree of freedom F_i and market minimum size $\underline{F}^{market}$: $F_i \geq \underline{F}^{market}$ for direct market eligibility.

Under aggregation: $\sum_{i \in \mathcal{A}} F_i \geq \underline{F}^{market}$.

Aggregation thus converts sub-threshold individual potentials into marketable aggregate volumes.

I.7.4 Liquidity

A general liquidity measure is $L = \mathcal{L}(N, \bar{F}, \kappa, \phi)$, with:

- N : number of participants,
- $\bar{F}$: average volume,
- κ : product compatibility,
- ϕ : friction measure.

Qualitatively: $\frac{\partial L}{\partial N} > 0$, $\frac{\partial L}{\partial \bar{F}} > 0$, $\frac{\partial L}{\partial \kappa} > 0$, $\frac{\partial L}{\partial \phi} < 0$. Liquidity is a structural market condition rather than a comfort variable.

I.7.5 Market tightness

With demand Q_t^D and available supply volume Q_t^S : $\chi_t = \frac{Q_t^D}{Q_t^S}$. The higher χ_t , the more relevant become:

- market tightness,
- price volatility,
- strategic bidding behaviour, and
- susceptibility to market power.

I.7.6 Friction costs

With friction costs C_i^{fric} : $U_i = E[R_i^{flex}] - C_i^{entry} - C_i^{oper} - C_i^{fric} - C_i^{risk}$. Two markets with identical revenue logic can therefore produce very different participation levels if their friction structures differ.

I.7.7 Product compatibility

With technical degree of freedom F_i^{tech} and marketable share F_i^{market} : $F_i^{market} = \kappa_i F_i^{tech}$, $0 \leq \kappa_i \leq 1$. κ_i depends on:

- response speed,
- minimum duration,
- minimum size,
- reservation capability,
- measurement and validation capability, and
- balancing and contractual compatibility.

Sound market design raises κ_i for a broad participant base.

I.7.8 Minimum condition for robust market formation

With total marketable volume $Q^{market} = \sum_{i \in J} F_i^{market}$, a minimum condition for robust market formation is:
 $Q^{market} \geq Q^{robust}$.

This threshold is context-dependent but conceptually central, for local markets.

I.7.9 Bidding behaviour

For supplier i with bid b_i : $\max_{b_i} E[\Pi_i(b_i, b_{-i})]$ with $\Pi_i = p_i(b_i, b_{-i}) \cdot (\pi_i^{award} - c_i^{true}) \cdot F_i$. In liquid markets, bids tend to approach opportunity costs. In thin markets, strategic elements gain in weight.

I.7.10 Nash logic

A bid profile $(b_1^*, \dots, b_N^*)$ is a Nash equilibrium when: $\Pi_i(b_i^*, b_{-i}^*) \geq \Pi_i(b_i, b_{-i}^*) \quad \forall b_i$. The significance here lies less in an explicit equilibrium solution than in the observation that market participants respond to one another and that market liquidity shapes strategic behaviour.

I.7.11 Bilevel and Stackelberg-type structures

Market design and participant behaviour are frequently hierarchically coupled. Formally:

Upper level: $\max_y W(y, x^*(y))$

Lower level: $x^*(y) = \arg \max_x \Pi(x; y)$.

Product definitions, minimum sizes, pricing rules, and activation logic thereby actively shape the market rather than merely observing it.

I.7.12 Regulatory parameters

With regulatory parameter ρ : $U_i(\rho) = E[R_i^{flex}(\rho)] - C_i^{entry}(\rho) - C_i^{oper}(\rho) - C_i^{risk}(\rho)$.

Regulation thus acts directly on

- participation,
- product compatibility, f
- frictions,
- liquidity, and
- ultimately on the visibility of technical degrees of freedom in the market.

I.7.13 Platforms

Platforms reduce frictions and raise compatibility:

$$\phi^{platform} < \phi^{no platform}, \quad \kappa^{platform} > \kappa^{no platform}.$$

From which it follows typically that: $L^{platform} > L^{no platform}$.

Platforms create no physical flexibility but raise the proportion of technical flexibility that becomes marketable.

I.7.14 Market quality

A general market quality measure can be written as: $MQ = \mathcal{M}(L, \kappa, \phi, \rho, \Gamma, \Omega)$, with:

- L : liquidity,
- κ : product compatibility,
- ϕ : frictions,
- ρ : regulatory parameters,
- Γ : baseline, validation, and settlement quality,
- Ω : susceptibility to strategic distortion or market power.

A flexibility market is therefore to be assessed not through product logic alone but through the coupling of technology, participation incentives, governance, and institutional robustness.

I.7.15 Structural conclusion

Incentives, liquidity, and market design determine the threshold between technically present and economically effective flexibility.

Asset models describe what is physically possible.

Market and dispatch models describe how degrees of freedom can be used.

Portfolio and hedging mathematics describe how risks and revenue spaces are brought together.

Baseline, validation, and settlement make performance institutionally robust.

Incentive and liquidity logic ultimately determines whether this entire structure translates into actual market participation.

Annex II - Acronyms

II.1 Core terms from energy, markets, and flexibility

Acronym	Name	Definition / classification
aFRR	automatic Frequency Restoration Reserve	Automatically activated balancing reserve used to restore the active power and frequency balance of the system.
BESS	Battery Energy Storage System	Electrochemical energy storage system comprising cells, power electronics, control, protection, and thermal management.
BKV	Balance Responsible Party	Market role responsible for maintaining the commercial balance of a balancing group.
CfD	Contract for Difference	Price-hedging arrangement under which the difference between a reference price and the market price is financially settled.
COP	Coefficient of Performance	Performance ratio of a heat pump; the ratio of useful heat output to electrical power input.
CPPA	Corporate Power Purchase Agreement	Long-term electricity offtake agreement with an industrial or commercial end customer.
DoD	Depth of Discharge	Degree of discharge of a storage system relative to its usable capacity.
DSM	Demand Side Management	Organisational and technical control framework for the targeted management of demand behaviour.
DSF	Demand Side Flexibility	Actually utilisable flexibility potential on the demand side.
EaaS	Energy as a Service	Service-oriented model in which energy and flexibility services are provided as an integrated solution.
EMS	Energy Management System	System used to monitor, coordinate, and optimise energy-related states and decisions.
ETRM	Energy Trading and Risk Management	Organisational and system-based backbone for trading, position management, risk monitoring, and settlement of energy portfolios.
EV	Electric Vehicle	Electrically powered vehicle.
FCR	Frequency Containment Reserve	Balancing reserve used for the immediate stabilisation of system frequency.
HP	Heat Pump	Heat pump.
H ₂	Hydrogen	Hydrogen.
ID	Intraday	Short-term electricity market used to adjust positions after the day-ahead market and up to close to the point of delivery.
IPP	Independent Power Producer	Independent electricity producer with its own or controlled generation base and responsibility for route-to-market activities.
LFP	Lithium Iron Phosphate	Lithium iron phosphate chemistry with high thermal stability and good cycle life.
LTO	Lithium Titanate Oxide	Lithium titanate chemistry with high power capability and strong cycle stability.
mFRR	manual Frequency Restoration Reserve	Manually activated balancing reserve used to restore the active power and frequency balance of the system.
MW	Megawatt	Unit of power.
MWh	Megawatt-hour	Unit of energy.
NMC	Nickel Manganese Cobalt	Lithium-ion chemistry with comparatively high energy density.
PCS	Power Conversion System	Power-electronic unit for conversion between direct current and alternating current.
PEM	Proton Exchange Membrane	Electrolysis technology with high dynamic response and strong suitability for variable power environments.
PPA	Power Purchase Agreement	Long-term electricity offtake agreement between a generator and an offtaker or route-to-market partner.
PV	Photovoltaics	Direct conversion of solar radiation into electrical power.
SoC	State of Charge	Current charge level of a storage system relative to its usable capacity.
SoH	State of Health	Measure of the technical condition and ageing state of an energy storage system.
SOEC	Solid Oxide Electrolysis Cell	High-temperature electrolysis technology with a distinct efficiency and integration profile.
TSO	Transmission System Operator	Transmission system operator.
VPP	Virtual Power Plant	Digital and operational aggregation of multiple distributed assets into a controllable unit.

II.2 Market and contract definitions

Term	Definition / classification
Aggregator	Market participant that aggregates the flexibility potential of multiple assets or market actors into a marketable volume
Balancing Energy	Energy and associated cost logic used to settle deviations between the scheduled and actual behaviour of a balancing group
Baseline	Reference value against which a flexibility activation is measured and assessed
Balancing Group	Commercially defined set of injections and withdrawals assigned to a specific responsibility area
Balance Responsible Party	Market role responsible for maintaining the balance of a balancing group
Curtailment	Deliberate reduction of otherwise available generation, usually for network or market-related reasons
Day-Ahead Market	Wholesale electricity market for delivery on the following day
Direct Marketing	Marketing of electricity from generation assets directly into the market or via a route-to-market partner
Congestion Management	Measures aimed at preventing or alleviating network congestion
Flexibility Market	Market mechanism in which the capability to modify generation, consumption, storage, or conversion in a targeted manner is economically valued
Independent Aggregator	Aggregator operating independently of a traditional energy supplier or other incumbent market roles
Imbalance Risk	Risk arising from deviations between the scheduled and actual behaviour of a portfolio or balancing group
Intraday Market	Market for short-term adjustment of electricity positions after closure of the day-ahead market
Local Flexibility Market	Flexibility market whose value logic depends materially on the location of activation within the network
Market Coupling	Coordinated integration of national or bidding-zone electricity markets through shared pricing and allocation mechanisms
Balancing Services	Standardised flexibility market used to maintain active power and frequency balance in the electricity system
Redispatch	Targeted adjustment of generation or demand patterns in order to influence power flows and relieve network congestion
Route-to-Market	Overall commercial pathway through which an asset or portfolio is integrated into market and contractual structures
Transfer of Energy	Commercial and procedural allocation of the energy shift resulting from a flexibility activation
Vertical Flexibility Market	Coordination mechanism linking multiple layers of network, market, balancing-group, and platform logic
Value Stacking	Combination of multiple use values or system contributions of an asset or portfolio
Revenue Stacking	Layering of multiple revenue streams from different markets, products, or contractual arrangements

II.3 Asset, technology, and system definitions

Term	Definition / classification
Asset	Physical unit or functionally integrated group of installations used for the generation, storage, conversion, or consumption of energy
Co-location	Spatial or network-side integration of multiple assets at one site or under a common grid connection point
Degradation	Gradual ageing of a technical system, in particular a storage system, due to time, temperature, operational stress, or cycling
Flexibility Potential	Actually utilisable operating range within which an asset or portfolio can deliberately modify its behaviour
Hybrid Portfolio	Combination of different asset classes whose interaction creates additional flexibility potential or more robust value and risk structures
Observability	Sufficiently accurate observability of a system or asset to derive robust state and control information
Portfolio	Commercial, operational, and route-to-market aggregation of multiple assets within a shared decision and control space
Round-Trip Efficiency	Efficiency of a storage system over a full charge and discharge cycle
Sector Coupling	Systematic integration of the electricity sector with heat, mobility, industry, or gaseous energy carriers
State of Charge	Current charge level of a storage system relative to its usable capacity
State of Health	Measure of the ageing condition and technical health of a technical system, in particular a storage system
Telemetry	Continuous technical acquisition and transmission of condition and operating data
Virtual Power Plant	Operational and digital integration architecture that combines multiple distributed assets into a controllable unit
Heat Pump	Electrically driven system used to raise thermal energy to a usable temperature level
Hydrogen Pathway	End-to-end technical and operational pathway from electricity intake via electrolysis and storage through to the use or further processing of hydrogen

II.4 Operational, risk, and portfolio definitions

Term	Definition / classification
Congestion Risk	Risk that network congestion may impair planned route-to-market or dispatch logic, or create additional local opportunities
Dispatch	Operational control or activation of assets over time
Forecasting	Model-based prediction of generation, demand, prices, system states, or network conditions
Hedging	Deliberate structuring or limitation of economic risks through physical, contractual, or financial offsetting positions
KPI Logic	Structured selection of performance and control indicators for the operational and commercial management of a portfolio
Operating Model	Functional architecture of roles, processes, decision rights, and systems for managing a portfolio or business model
Optimisation Portfolio	Portfolio whose economic value arises from the coordinated management of multiple flexibility potentials and revenue pools
Scheduling	Temporal derivation and arrangement of planned operating profiles, positions, or activation windows
Settlement	Financial and commercial settlement of services delivered, including allocation, validation, and remuneration
Shape Risk	Risk that the temporal profile of an asset or portfolio does not align with price, delivery, or obligation profiles
Participation Incentive	Economic and organisational attractiveness of making a flexibility potential available in a market or mechanism
Validation	Assessment of whether a measured change can be recognised as flexibility service actually delivered

Annex III - References

III.1 Regulatory, institutional, and market-related documents

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